System: 系统	**ABS 9.x** **ESP® 9.x** **ESP®plus 9.x**	**Equipment: 设备**	**Bleeding device 排气设备**
		Diagnostic interface: 诊断接口	**K-Line (opt.); CAN;** **FlexRay (customer specific)**
Brake circuit configuration:	**Diagonal (X)** **Front / Rear (II)**	**Diagnostic protocol: 诊断协议**	**Customer specific 客户规范**
Customer:	**General**		
"D"			
Distribution:	Customer		

	Name	Signature:	Date:
Checked by:	Jiang zhigang		
Approved by:	Przybyla Benjamin		

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1 General 综述

1.1 Acronyms 缩写

ABS	antilock braking system	防抱死系统
ABS-SILA	ABS warning light	ABS报警灯
AC	Accumulator chamber	蓄能仓
AV	outlet valve	出液阀
ALS	longitudinal acceleration sensor	纵向加速度传感器
AYS	lateral acceleration sensor	横向加速度传感器
BLS	brake light switch	制动灯开关
BS	Brake switch	制动开关
DTC	diagnostic trouble code	诊断故障码
EBD	electronic brake distribution	电子制动力分配
ECU	electronic control unit	电子控制单元
ESP	electronic stability program	电子稳定程序
ESP-MFL	ESP multifunction (warning) light	ESP多功能报警灯
EV	inlet valve	进液阀
FA	front axle	前轴
FCM	fault code memory	故障码存储器
FL	front left	前左
FR	front right	前右
HAS	handbrake switch (park brake switch)	手刹信号开关
HSV	high pressure switching valve(priming valve)	高压开关阀
MCC	master cylinder circuit	主缸回路
PATA	passive taster (TCS/DSC passive taster)	ESP开关
PB	process byte	过程字节
PS (DG, DS)	pressure sensor	压力传感器
RA	rear axle	后轴
RFP	recirculation pump	回流泵
RL	rear left	后左
RR	rear right	后右
SAS (LWS)	steering angle sensor	转向角传感器
SV	solenoid valve	电磁阀
UMR	pump motor feedback	泵电机反馈
USV	pilot valve	先导阀
UVR	Valve relay feedback	阀继电器
WSS	wheel speed sensor	轮速传感器
YRS	yaw rate sensor	横摆角速度传感器

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2 General 总则

This document describes each step for a repair bleeding process of an ABS or ESP system. This procedure respectively repair bleeding process has to carry out in case of the vacuum process is failed or a dry ABS/ESP hydraulic unit is assembled in filled vehicle.
该文件描述了ABS或ESP系统每个排气过程步骤。对在抽真空/加注线上失败的ABS/ESP或者干式ABS/ESP进行离线加注时，必须根据本文件描述的步骤进行人工排气。

2.1 Scope 范围

All the Bosch ABS and ESP components, such as hydraulic modulator are delivered to the customer assembly plants in a dry state (free of brake fluid), that means that the primary and secondary circuits of the hydraulic modulator are complete dry.
To bleed respectively fill the secondary circuits of hydraulic modulator different valves must be energized.
At vehicles which fail the online evacuation and filling process or in case of replacement of the hydraulic modulator by a dry ABS / ESP unit a special repair bleeding procedure has to be carry out.
During execution of this repair bleeding procedure different valves and pump motor have to be activated in an exact defined sequence and the operator has to open the bleeding screw at the respective wheel.
During the repair bleed process the brake fluid is pressed by the pressure of the bleeding device through the brake fluid reservoir, master cylinder, the entire brake system and the respective opened bleeding screw. In addition the brake pedal has to be actuated to support the bleeding process.
The execution of all required valve and pump motor activations in the defined sequence is realised by a diagnostic tester. This tester is connected with the ABS / ESP ECU via the diagnostic interface (OBD2) within the passenger cell (precondition: all components such as wiring harness and battery complete assembled).
The bleeding sequence is done in the same way as at a vehicle with a standard brake system without an ABS or ESP unit with exception of the additional time which is required to bleed the secondary and priming (only ESP) circuits of the ABS / ESP hydraulic unit.
The used bleeding device has to fulfill the requirements which are specified in this document.
所有的博世 ABS 和 ESP 部件，如液压调节器是以干燥的状态（无制动液）提交到客户装配工厂的，液压调节器的第一回路和第二回路都是干燥的。
对液压模块的第二回路进行排气和加注时不同的阀必须被激活。
所有在线抽真空/加注过程中失败，或者更换干式件的系统，都必须以本文描述的特定程序进行人工排气。
在排气过程中不同的阀和马达都将有序的激活，同时操作者须打开相应的排气螺母。
排气设备在排气过程中对制动液施压，并使液体通过储液壶，主缸及整个制动系统从相应打开的排气螺母排出。
同时在整个过程中必须往复地踩刹车踏板。
所有阀和马达有序的激活都是由诊断设备来实现的。在乘客舱内诊断设备都是通过（OBD2）诊断口来连接ABS/ESP电控单元。（前提：线束和电瓶这样的零部件必须事先完成安装）
这个过程和标准的无 ABS/ESP 制动系统排气基本一样，除了在时间上有一定的延长，这是为了满足液压调节器第二回路的排气要求。
用于排气的设备必须满足该文件定义的要求。

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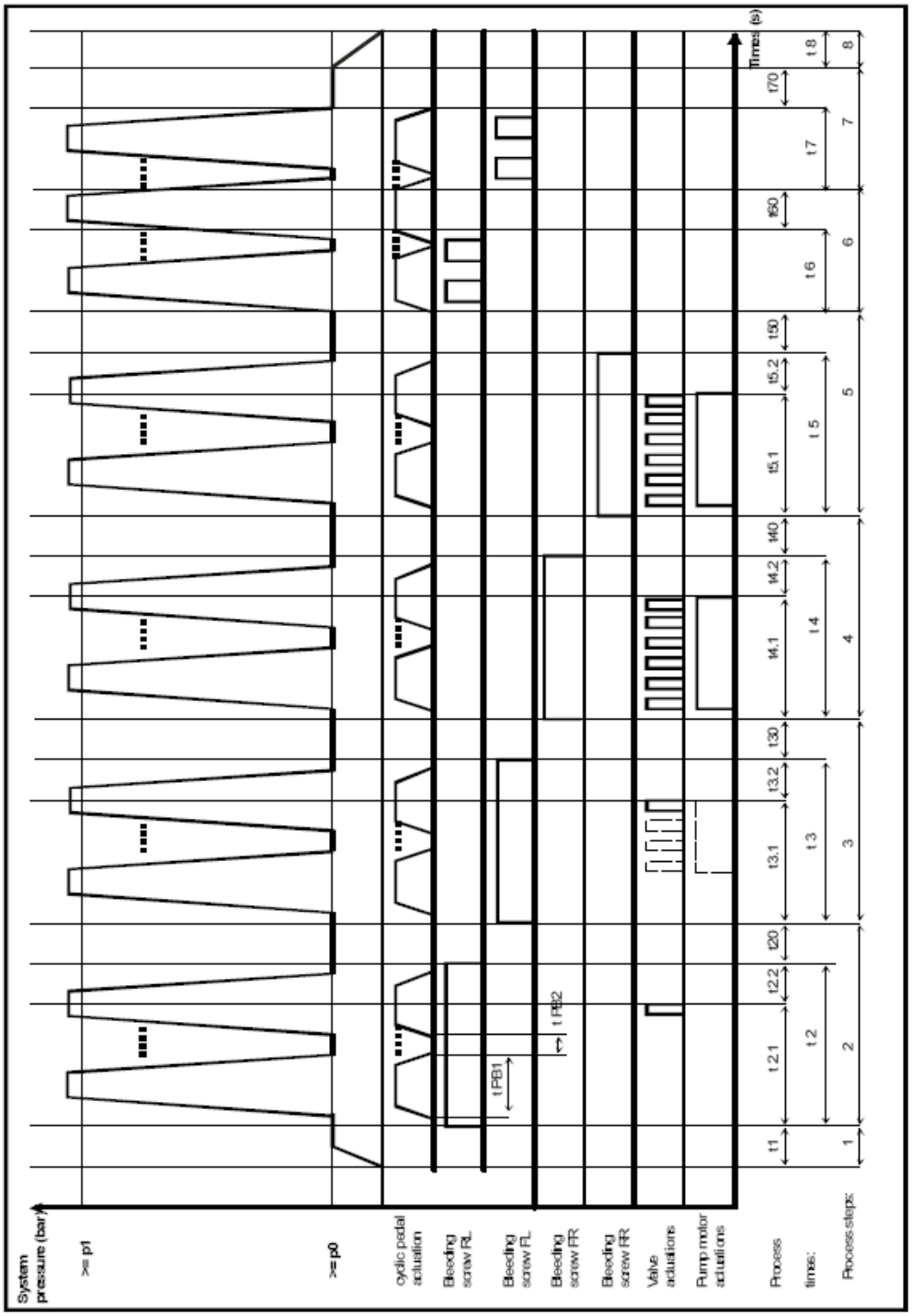
2.2 Process sequence 过程顺序

Diagram of the process sequence: 过程顺序图



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Explanations to the process sequence diagram: 过程顺序图解

Process step:	Time: 时间	Process description: 过程描述	
1	t1	At the beginning of the process the bleeding device and the electric connections have to be adapted. To guarantee a good bleeding result, a defined **absolute pressure p_0** at the bleeding device must be adjusted and the brake pedal must be cyclic actuated during the whole bleeding process. By the **cyclic brake actuation** is produced an appropriate **system pressure p_1** which supports the bleeding process considerably. In dependence of the accessibility of the bleeding screws the vehicle must be raised if necessary and all wheels be dismounted. The required **time t_1** for the **handling 1** is dependent on the operator respectively the conditions at the vehicle. 在排气前必须完成排气设备和电气接头的连接。为确保排气成功在整个排气过程中必须保证排气设备调整到绝对压力**p_0**且始终反复踩踏制动踏板。由制动回路建立的系统压力**p_1**将支持排气过程。为能方便的接触到排气螺母必须将车辆升起，必要的话可以卸下所有车轮。**操作1**所需的**时间t_1**取决于人员和车辆情况。	
2	t2	t2.1	The whole **repair bleeding process for the wheel RL** for the **bleeding time t_2** consists of the **phase duration $t_{2.1}$** and **waiting period $t_{2.2}$**. After starting the associated bleeding time t_x the appropriate wheel circuit is bleed. In parallel the associated valve controls (phase duration $t_{x.x}$) is started. During this time the specified valve activations are executed for the appropriate wheel to bleed the secondary circuit in the ABS / ESP hydraulic unit. **后左轮整个排气过程的排气时间t_2**包括了阶段持续时间**$t_{2.1}$**和等候时间**$t_{2.2}$**。在开始排气时间t_x后相应车轮的回路排气完成，同时指定的阀控制也开始执行(阶段持续t_{xx})。在此期间为执行ABS/ESP液压模块第二回路的排气，将对指定的阀进行激活。
		t2.2	After the specified valve controls (phase duration $t_{x.x}$) are finished, the appropriate wheel must be bleed as long as the brake fluid is seen in the bleeding tube (connected to the bleeding screw) free of air and bubbles. This required waiting period $t_{x.x}$ depends on the vehicle, the design and size of the base brake system. This time is to be specified by the customer. 在指定的阀控制(阶段持续t_{xx})结束后，必须确保相应轮排气的完成，即从制动管出来的制动液（与排气螺母相连）必须无气泡。等待阶段的时间t_{xx}取决于车辆状况和制动系统的设计、尺寸。该时间由客户定义。
	t20	The **handling time t_{20}** is a time dependent on the operator and it is required to connect the bleeding bottle respectively opening and closing of the bleeding screws. 操作时间**t_{20}**取决于操作者连接排气头和打开锁紧排气螺母的时间。	
3	t3	t3.1	The whole **repair bleeding process for the wheel FL** for the **bleeding time t_3** consists of the **phase duration $t_{3.1}$** and **waiting period $t_{3.2}$**. For more details see the description of the process step 2. **前左轮整个排气过程的排气时间t_3**包括了阶段持续时间**$t_{3.1}$**和等候时间**$t_{3.2}$**。详细请参考过程描述步骤2。
		t3.2	
	t30	The **handling time t_{30}** is a time dependent on the operator. For more details see the description of the process step 2. 操作时间**t_{30}**取决于操作者。详细请参考过程描述步骤2。	

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4	t4	t4.1 t4.2	The whole **repair bleeding process for the wheel FR** for the **bleeding time t_4** consists of the **phase duration $t_{4.1}$** and **waiting period $t_{4.2}$**. For more details see the description of the process step 2. **前右轮整个排气过程的排气时间t_4**包括了阶段持续时间**$t_{4.1}$**和等候时间**$t_{4.2}$**。详细请参考过程描述步骤2。
	t40	The **handling time t_{40}** is a time dependent on the operator. For more details see the description of the process step 2. 操作时间**t_{40}**取决于操作者。详细请参考过程描述步骤2。	
5	t5	t5.1 t5.2	The whole **repair bleeding process for the wheel RR** for the **bleeding time t_5** consists of the **phase duration $t_{5.1}$** and **waiting period $t_{5.2}$**. For more details see the description of the process step 2. **后右轮整个排气过程的排气时间t_5**包括了阶段持续时间**$t_{5.1}$**和等候时间**$t_{5.2}$**。详细请参考过程描述步骤2。
	t50	The **handling time t_{50}** is a time dependent on the operator. For more details see the description of the process step 2. 操作时间**t_{50}**取决于操作者。详细请参考过程描述步骤2。	
6	t6	In dependence of the brake split configuration and the used orifices in the ABS / ESP hydraulic unit the **wheel RL** must be bleed in the individual case again for the **bleeding time t_6 without valve controls**. **This step is optional.** The necessity of this additional bleeding step must be determined by attempts executed with vehicles with completely dry brake system. The responsibility for these trials has the customer. 根据不同的制动回路分布类型（X型或II型）与ABS/ESP液压单元截流口尺寸，个别情况下**后左轮**必须在**阀不启动时**再排一次，排气时间为**t_6**。作为可选项，该额外的排气步骤是否执行应当通过试验来确定。这些试验应由客户在安有全干式制动系统的汽车上进行。	
	t60	The **handling time t_{60}** is a time dependent on the operator. For more details see the description of the process step 2. 操作时间**t_{60}**取决于操作者。详细请参考过程描述步骤2。	
7	t7	In dependence of the brake split configuration and the used orifices in the ABS / ESP hydraulic unit the **wheel FL** must be bleed in the individual case again for the **bleeding time t_7 without valve controls**. **This step is optional.** The necessity of this additional bleeding step must be determined by attempts executed with vehicles with completely dry brake system. The responsibility for these trials has the customer. 根据不同的制动回路分布类型（X型或II型）与ABS/ESP液压单元截流口尺寸，个别情况下**前左轮**必须在**阀不启动时**再排一次，排气时间为**t_7**。作为可选项，该额外的排气步骤是否执行应当通过试验来确定。这些试验应由客户在安有全干式制动系统的汽车上进行。	
	t70	The **handling time t_{70}** is a time dependent on the operator. For more details see the description of the process step 2. 操作时间**t_{70}**取决于操作者。详细请参考过程描述步骤2。	
8	t8	At the end of the process the bleeding device and the electric connections have to be disconnected respectively the pedal stroking have to be stopped. The required **time t_8** for the **handling 2** is dependent on the operator respectively the conditions at the vehicle. 在过程结束时必须分别断开排气设备和电气接头，停止踩制动踏板。操作2所需的时间**t_8**取决于操作者和车辆情况。	

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3 Description of each process step 单个过程步骤描述

3.1 Process step 1: Handling 1 过程步骤 1：操作 1

Handling 1 操作1

Operator: 操作者

☞ Adapt the bleeding device with the adjusted absolute pressure **p_0** at the brake fluid reservoir
连接排气设备至制动液储液壶并调整到绝对压力**p_0**

☞ Adapt the electrical connections via the diagnostic interface (OBD2) in the vehicle
将电气接头经诊断接口（OBD2）连接至车上

☞ Switch the ignition on
点火

☞ Check the accessibility of the bleeding screws, if necessary raise the vehicle and dismount the wheels
检查排气螺母，必要的话升起车辆并卸下车轮

Process step:	Parameter: 参数	Physical values:
1	Absolute pressure p0 绝对压力p0	≥ y bar
	Handling time t1 操作时间t1	x s

Notes: 注意

In the plant rework area the vehicles are completely assembled. The electrical connection respectively the diagnostics are done via the OBD2 connector within the passenger cell. The operator has to switch on the ignition.
To support the bleeding process a bleeding device with the adjusted absolute pressure p0 has to be adapted at the brake fluid reservoir.
The bleeding sequence of the wheels has not been changed in comparison to the Bosch generation 8. Following bleeding sequence has to be guaranteed; otherwise residual air can remain back in the brake system:
车辆进入工厂返工区域前都已组装完成。经OBD2通过和车辆的诊断连接，完成电气连接。
操作者须打开点火开关。
排气过程要求排气设备在制动液储液壶处的绝对压力须调整到p0。
车轮排气的先后顺序较博世8代的产品没有任何改变。
必须保证以下排气的先后顺序，否则残留空气还会在制动管路中存在。

1. Rear left (RL) 后左
2. Front left (FL) 前左

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3. Front right (FR) 前右
4. Rear right (RR) 后右
5. Rear left (RL) - optional 后左 – 可选
6. Front left (FL) - optional 前左 – 可选

The necessity to execute these optional process steps 5 & 6 are dependent on following conditions:
过程步骤5和6是否有必要执行取决于以下情形：

. • Brake split configuration 制动回路分布类型
. • Used orifices in the ABS/ESP hydraulic unit ABS/ESP液压单元截流口尺寸
. • Filling state of the base brake system (partially or completely filled) 基础制动系统加注状态（部分或全部加注）

Regarding the bleed ability of a complete brake system, vehicles with a front/rear brake split configuration and an ABS/ESP system with small inlet orifices on the rear axle wheels are the worst case.
就整个制动系统的排气能力而言，车辆制动回路分布为II型且ABS/ESP系统的后轴进液截流口较小时，排气能力是最差的。

To find out if the optional bleeding steps 5 & 6 are really necessary, corresponding trials have to be executed by the customer.
为了判断可选的排气步骤5、6是否必要，必须由客户通过相应试验来验证。

The with x marked **handling time t1** depends on the installation position of the brake fluid reservoir, the position of the diagnosis interface OBD2 in the vehicle interior and the conditions at the vehicle. The handling time t1 must be determined together with the operators by trials. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的**操作时间t1**取决于制动液储液壶安装位置、诊断接口OBD2在车内的位置以及车辆状况。同时，操作时间**t1**还必须经过操作者试验确认。这些试验应由客户自行完成。

All with y marked **absolute pressure p0** is to be inferred from the chapter **Process data (pressure and time values)**.
数值为y的**绝对压力p0**可由“过程数据（压力与时间值）”相关章节得出。

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3.2 Process step 2: Bleeding of the wheel circuit rear left (RL) 过程步骤 2：后左轮回路排气

Bleeding of the wheel circuit rear left (RL) 后左轮回路排气	
Operator: 操作者	Diagnostic unit / ECU: 诊断单元/电子控制单元
☞Connect the bleeding bottle and open the bleeding screw at the wheel **RL**. 连接排气头并打开后左轮的排气螺母。 ☞Actuate the brake pedal alternately with a normal speed (actuated for the time t_{BP1}; released for the time t_{BP2}). This step could also automated by a pneumatic or electrical pedal actuation tool. 以正常的速度反复踩制动踏板（踩住时间t_{BP1}，松开时间t_{BP2}）。该步骤也可以由气动或电动的自动踩踏工具来实现。 ☞Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed. 通过诊断设备开始诊断程序（如阀控制等），等待直至显示程序结束。 ☞Close the bleeding screw at the wheel **RL** and disconnect the bleeding bottle. 锁紧后左轮的排气螺母并断开排气头。 ☞Stop the brake pedal actuation. 停止踩制动踏板。	Specified valve activations plus the defined waiting time for the wheel **RL** are executed. 对后左轮执行指定的阀控制和等待时间。

Process step:	**Parameter: 参数**		**Physical values: 重要的值**
2	**Bleeding time t_2** (consist of the diagnostic times): 排气时间t_2（包括诊断时间）	phase duration $t_{2.1}$ 阶段持续时间$t_{2.1}$	**y s**
		waiting period $t_{2.2}$ 等待时间$t_{2.2}$	**x s**
	System pressure p1 系统压力p1		**y bar**
	Time t_{BP1} 时间t_{BP1}		**y s**
	Time t_{BP2} 时间t_{BP2}		**y s**
	Handling time t_{20} 操作时间t_{20}		**x s**

Notes: 注意：

The **bleeding screw RL** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder

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respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time t_{BP1}**; released for the **time t_{BP2}**). This brake pedal stroking creates the **system pressure p1**, which supports the bleeding process significantly. Analogously the **specified valve activations** for the **wheel RL** are executed for the **phase duration $t_{2.1}$** to bleed the secondary circuit of the ABS/ESP hydraulic unit.
除了后左轮其他轮的排气螺母都为锁紧状态。制动液依次通过ABS/ESP液压单元，制动管，轮缸，卡钳和排气螺母被排气设备排出。在过程中制动踏板必须反复踩踏（踩住时间t_{BP1}；松开时间t_{BP2}）。制动踏板提供的**系统压力p1**保证了排气过程的进行。同时在持续阶段$t_{2.1}$中，后左轮指定的阀开始动作，以对ABS/ESP液压单元第二回路进行排气。

After the specified valve activations are finished, the wheel RL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period $t_{2.2}$** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t_2** for the wheel RL consists of the phase duration $t_{2.1}$ and the waiting period $t_{2.2}$. The **handling time t_{20}** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.
在指定的阀动作结束后，必须确保后左轮的排气完成，即在排气螺母口出来的制动液必须无气泡。因此额外的等待阶段时间$t_{2.2}$取决于车辆状况和基础制动系统设计。该时间由客户定义。后左轮总的排气时间**t_2**包括了持续阶段$t_{2.1}$和等待阶段$t_{2.2}$。操作时间t_{20}取决于操作者连接排气头和打开锁紧排气螺母的时间。

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的时间值取决于制动系统的尺寸，排气压力和车辆的具体情况（如操作排气螺母的便捷性等）。该时间必须在客户装配工厂通过试验获得。操作时间也必须通过操作者试验确定。这些试验应由客户自行完成。

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.
数值为y的压力和时间值可由“过程数据（压力和时间值）”和“诊断数据（时间值）”相关章节得出。

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3.3 Process step 3: Bleeding of the wheel circuit front left (FL) 过程步骤 3：前左轮回路排气

Bleeding of the wheel circuit front left (FL) 前左轮回路排气	
Operator: 操作者	Diagnostic unit / ECU: 诊断单元/电子控制单元
☞Connect the bleeding bottle and open the bleeding screw at the wheel **FL**. 连接排气头并打开前左轮的排气螺母。 ☞Actuate the brake pedal alternately with a normal speed (actuated for the time t_{BP1}; released for the time t_{BP2}). This step could also automated by a pneumatic or electrical pedal actuation tool. 以正常的速度反复踩制动踏板（踩住时间t_{BP1}，松开时间t_{BP2}）。该步骤也可以由气动或电动的自动踩踏工具来实现。 ☞Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed. 通过诊断设备开始诊断程序（如阀控制等），等待直至显示程序结束。 ☞Close the bleeding screw at the wheel **FL** and disconnect the bleeding bottle. 锁紧前左轮的排气螺母并断开排气头。 ☞Stop the brake pedal actuation. 停止踩制动踏板。	Specified valve activations plus the defined waiting time for the wheel **FL** are executed. 对前左轮执行指定的阀控制和等待时间。

Process step:	**Parameter: 参数**		**Physical values: 重要的值**
3	**Bleeding time t3** (consist of the diagnostic times): 排气时间t3（包括诊断时间）	phase duration $t_{3.1}$ 阶段持续时间$t_{3.1}$	**y s**
		waiting period $t_{3.2}$ 等待时间$t_{3.2}$	**x s**
	System pressure p1系统压力p1		**y bar**
	Time t_{BP1} 时间t_{BP1}		**y s**
	Time t_{BP2} 时间t_{BP2}		**y s**
	Handling time t_{30} 操作时间t_{30}		**x s**

Notes: 注意：

The **bleeding screw FL** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder

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respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time tBP1**; released for the **time tBP2**). This brake pedal stroking creates the **system pressure p1**, which supports the bleeding process significantly. Analogously the **specified valve activations** for the **wheel FL** are executed for the **phase duration 3.1** to bleed the secondary circuit of the ABS/ESP hydraulic unit.
除了前左轮其他轮的排气螺母都为锁紧状态。制动液分别通过ABS/ESP液压单元，制动管，轮缸，卡钳和排气螺母被排气设备排出。在过程中制动踏板必须反复踩踏（踩住时间t_{BP1}；松开时间t_{BP2}）。制动踏板提供的**系统压力p1**保证了排气过程的进行。同时在持续阶段$t_{3.1}$中，前左轮指定的阀开始动作，以对ABS/ESP液压单元第二回路进行排气。

After the specified valve activations are finished, the wheel FL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period t3.2** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t3** for the wheel FL consists of the phase duration t3.1 and the waiting period t3.2. The **handling time t30** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.
在指定的阀动作结束后，必须确保前左轮的排气完成，即在排气螺母口出来的制动液必须无气泡。因此额外的等待阶段时间$t_{3.2}$取决于车辆状况和基础制动系统设计。该时间由客户定义。前左轮总的排气时间**t_3**包括了持续阶段$t_{3.1}$和等待阶段$t_{3.2}$。操作时间t_{30}取决于操作者连接排气头和打开锁紧排气螺母的时间。

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的时间值取决于制动系统的尺寸，排气压力和车辆的具体情况（如操作排气螺母的便捷性等）。该时间必须在客户装配工厂通过试验获得。操作时间也必须通过操作者试验确定。这些试验应由客户自行完成。

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values)**.
数值为y的压力和时间值可由“过程数据（压力和时间值）”和“诊断数据（时间值）”相关章节得出。

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3.4 Process step 4: Bleeding of the wheel circuit front right (FR) 过程步骤 4：前右轮回路排气

Bleeding of the wheel circuit front right (FR) 前右轮回路排气	
Operator: 操作者	Diagnostic unit / ECU: 诊断单元/电子控制单元
☞Connect the bleeding bottle and open the bleeding screw at the wheel FR. 连接排气头并打开前右轮的排气螺母。 ☞Actuate the brake pedal alternately with a normal speed (actuated for the time tBP1; released for the time tBP2). This step could also automated by a pneumatic or electrical pedal actuation tool. 以正常的速度反复踩制动踏板（踩住时间t_{BP1}，松开时间t_{BP2}）。该步骤也可以由气动或电动的自动踩踏工具来实现。 ☞Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed. 通过诊断设备开始诊断程序（如阀控制等），等待直至显示程序结束。 ☞Close the bleeding screw at the wheel FR and disconnect the bleeding bottle. 锁紧前右轮的排气螺母并断开排气头。 ☞Stop the brake pedal actuation. 停止踩制动踏板。	Specified valve activations plus the defined waiting time for the wheel FR are executed. 对前右轮执行指定的阀控制和等待时间。

Process step:	**Parameter: 参数**		**Physical values: 重要的值**
4	**Bleeding time t4** (consist of the diagnostic times): 排气时间t_4（包括诊断时间）	Phase duration t4.1_ABS respectively. t4.1_ESP	**y s**
		waiting period t4.2 等待时间$t_{4.2}$	**x s**
	System pressure p1系统压力p1		**y bar**
	Time t_{BP1} 时间t_{BP1}		**y s**
	Time t_{BP2} 时间t_{BP2}		**y s**
	Handling time t_{40} 操作时间t_{40}		**x s**

Notes: 注意

The **bleeding screw FR** is opened all other bleeding screws are closed. The brake fluid is pushed out by the

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pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time tBP1**; released for the **time tBP2**). This brake pedal stroking creates the **system pressure p1**, which supports the bleeding process significantly.
除了**前右**轮其他轮的排气螺母都为锁紧状态。制动液分别通过ABS/ESP液压单元，制动管，轮缸，卡钳和排气螺母被排气设备排出。在过程中制动踏板必须反复踩踏（踩住时间t_{BP1}；松开时间t_{BP2}）。制动踏板提供的**系统压力p1**保证了排气过程的进行。

Analogously the **specified valve and pump motor activations** for the **wheel FR** are executed for the **phase duration t4.1_ABS** respectively **t4.1_ESP** to bleed the secondary circuit of the ABS/ESP hydraulic unit.
同时持续阶段$t_{4.1}$_ABS或t4.1_ESP中，**前右轮**指定的阀和泵马达开始动作，以对ABS/ESP液压单元第二回路进行排气。

After the specified valve activations are finished, the wheel FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period t4.2** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t4** for the wheel FR consists of the phase duration t4.1_ABS respectively t4.1_ESP and the waiting period t4.2. The **handling time t40** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.
在指定的阀动作结束后，必须确保**前右轮**的排气完成，即在排气螺母口出来的制动液必须无气泡。因此额外的等待阶段时间$t_{4.2}$取决于车辆和基础制动系统设计。该时间由客户定义。**前右**轮总的排气时间t_4包括了持续阶段$t_{4.1}$和等待阶段$t_{4.2}$。操作时间t_{40}取决于操作者连接排气头和打开锁紧排气螺母的时间。

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的时间值取决于制动系统的尺寸，排气压力和车辆的具体情况（如操作排气螺母的便捷性等）。该时间必须在客户装配工厂通过试验获得。操作时间也必须通过操作者试验确定。这些试验应由客户自行完成。

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values).**
数值为y的压力和时间值可由“过程数据（压力和时间值）”和“诊断数据（时间值）”相关章节得出。

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3.5 Process step 5: Bleeding of the wheel circuit rear right (RR) 过程步骤 5：后右轮回路排气

Bleeding of the wheel circuit rear right (RR) 后右轮回路排气	
Operator: 操作者	Diagnostic unit / ECU: 诊断单元/电子控制单元
☞Connect the bleeding bottle and open the bleeding screw at the wheel **RR**. 连接排气头并打开后右轮的排气螺母。 ☞Actuate the brake pedal alternately with a normal speed (actuated for the time tBP1; released for the time tBP2). This step could also automated by a pneumatic or electrical pedal actuation tool. 以正常的速度反复踩制动踏板（踩住时间t_{BP1}，松开时间t_{BP2}）。该步骤也可以由气动或电动的自动踩踏工具来实现。 ☞Start the diagnostics (valve activations, etc.) via the tester and wait till the end message is displayed. 通过诊断设备开始诊断程序（如阀控制等），等待直至显示程序结束。 ☞Close the bleeding screw at the wheel **RR** and disconnect the bleeding bottle. 锁紧后右轮的排气螺母并断开排气头。 ☞Stop the brake pedal actuation. 停止踩制动踏板。	Specified valve activations plus the defined waiting time for the wheel **RR** are executed. 对后右轮执行指定的阀控制和等待时间。

Process step:	**Parameter: 参数**		**Physical values: 重要的值**
5	**Bleeding time t5** (consist of the diagnostic times): 排气时间t_5（包括诊断时间）	phase duration t5.1_ABS respectively. t5.1_ESP	**y s**
		waiting period t5.2 等待时间$t_{5.2}$	**x s**
	System pressure p1系统压力p1		**y bar**
	Time t_{BP1} 时间t_{BP1}		**y s**
	Time t_{BP2} 时间t_{BP2}		**y s**
	Handling time t_{50} 操作时间t_{50}		**x s**

Notes: 注意

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The **bleeding screw RR** is opened all other bleeding screws are closed. The brake fluid is pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. To support this process **the brake pedal** has been **alternate actuated** (actuated for the **time tBP1**; released for the **time tBP2**). This brake pedal stroking creates the **system pressure p1**, which supports the bleeding process significantly.
除了**后右**轮其他轮的排气螺母都为锁紧状态。制动液分别通过ABS/ESP液压单元，制动管，轮缸，卡钳和排气螺母被排气设备排出。在过程中制动踏板必须反复踩踏（踩住时间t_{BP1}；松开时间t_{BP2}）。制动踏板提供的**系统压力p1**保证了排气过程的进行。

Analogously the **specified valve and pump motor activations** for the **wheel RR** are executed for the **phase duration t5.1_ABS** respectively **t5.1_ESP** to bleed the secondary circuit of the ABS/ESP hydraulic unit.
同时持续阶段t5.1_ABS或t5.1_ESP中，**后右轮**指定的阀和泵马达开始动作，以对ABS/ESP液压单元第二回路进行排气。

After the specified valve activations are finished, the wheel RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. The therefore needed additional **waiting period t5.2** depends on the vehicle respectively on the design of the base brake system. This time must be specified by the customer. The total **bleeding time t5** for the wheel RR consists of the phase duration t5.1_ABS respectively t5.1_ESP and the waiting period t5.2. The **handling time t50** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.
在指定的阀动作结束后，必须确保**后右**轮的排气完成，即在排气螺母口出来的制动液必须无气泡。因此额外的等待阶段时间$t_{5.2}$取决于车辆和基础制动系统尺寸的设计。该时间由客户定义。**后右**轮总的排气时间t_5包括了持续阶段$t_{5.1}$和等待阶段$t_{5.2}$。操作时间t_{50}取决于操作者连接排气头和打开锁紧排气螺母的时间。

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的时间值取决于制动系统的尺寸，排气压力和车辆的具体情况（如操作排气螺母的便捷性等）。该时间必须在客户装配工厂通过试验获得。操作时间也必须通过操作者试验确定。这些试验应由客户自行完成。

All with y marked **pressure and timing values** are to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values).**
数值为y的压力和时间值可由“过程数据（压力和时间值）”和“诊断数据（时间值）”相关章节得出。

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3.6 Process step 6: Bleeding of the wheel circuit rear left (RL) - optional 过程步骤 6：后左轮回路排气 - 可选

Bleeding of the wheel circuit rear left (RL) 后左轮回路排气	
Operator: 操作者	Diagnostic unit / ECU: 诊断单元/电子控制单元
☞Connect the bleeding bottle. 连接排气头 ☞Repeat following process more than one: 重复执行下列过程一次以上： • Actuate the brake pedal. 踩制动踏板 • Open the bleeding screw at the wheel RL. 打开后左轮排气螺母 • Close the bleeding screw at the wheel RL. 锁紧后左轮排气螺母 • Release the brake pedal. 松开制动踏板 ☞Disconnect the bleeding bottle. 断开排气头	---

Process step:	**Parameter: 参数**	**Physical values: 重要的值**
6	**System pressure p1系统压力p1**	**y bar**
	Bleeding time t6 排气时间t6	**x s**
	Handling time t60 操作时间t6	**x s**

Notes: 注意

To push out the residual air which is might remained back in the wheel brake circuit after the standard bleeding following procedure should be carry out: At first the bleeding screw at the wheel RL should be closed before the brake pedal is actuated. Then the brake pedal has to be powerful actuated to create a high system pressure. As next the bleeding screw has to be opened. Caused by the high system pressure the residual air is compressed and released within a short time after the bleeding screw was opened. The released air and the brake fluid are pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. The therefore required **bleeding time t6** has to be determined by trials at the customer plant. These trials are in responsibility of the customer and have to be executed by them self.
为了将标准排气过程之后仍然残留在轮缸制动回路中的空气排出，必须进行下列步骤：首先在制动踏板踩下之前锁紧后左轮的排气螺母，再用力踩下制动踏板以产生一个很高的制动系统压力，并随后打开下一个排气螺母。由于系统压力很高，残留空气将被压缩并在排气螺母打开后短时间内排出。残留空气连同制动液将依次通过

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ABS/ESP液压单元，制动管，轮缸，卡钳和排气螺母被排气设备排出。所需排气时间**t6**必须在客户工厂通过试验获得。这些试验应由客户自行完成。

The **handling time t60** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.
操作时间**t60**取决于操作者连接排气头和打开锁紧排气螺母的时间。

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的时间值取决于制动系统的尺寸，排气压力和车辆的具体情况（如操作排气螺母的便捷性等）。该时间必须在客户装配工厂通过试验获得。操作时间也必须通过操作者试验确定。这些试验应由客户自行完成。

The with y marked **pressure value** is to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values).**
数值为y的压力值可由“过程数据（压力和时间值）”和“诊断数据（时间值）”相关章节得出。

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3.7 Process step 7: Bleeding of the wheel circuit front left (FL) - optional 过程步骤 7：前左轮回路排气 - 可选

Bleeding of the wheel circuit front left (FL) 前左轮回路排气	
Operator: 操作者	Diagnostic unit / ECU: 诊断单元/电子控制单元
☞ Connect the bleeding bottle. 连接排气头 ☞ Repeat following process more than one: 重复执行下列过程一次以上： • Actuate the brake pedal. 踩制动踏板 • Open the bleeding screw at the wheel FL. 打开后左轮排气螺母 • Close the bleeding screw at the wheel FL. 锁紧后左轮排气螺母 • Release the brake pedal. 松开制动踏板 ☞ Disconnect the bleeding bottle. 断开排气头	---

Process step:	**Parameter: 参数**	**Physical values: 重要的值**
7	System pressure p1 系统压力p1	y bar
	Bleeding time t7 排气时间t7	x s
	Handling time t70 操作时间t70	x s

Notes: 注意

To push out the residual air which is might remained back in the wheel brake circuit after the standard bleeding following procedure should be carry out: At first the bleeding screw at the wheel FL should be closed before the brake pedal is actuated. Then the brake pedal has to be powerful actuated to create a high system pressure. As next the bleeding screw has to be opened. Caused by the high system pressure the residual air is compressed and released within a short time after the bleeding screw was opened. The released air and the brake fluid are pushed out by the pressure of the bleeding device through the ABS/ESP hydraulic unit, the brake pipe, wheel brake cylinder respectively calliper and the bleeding screw. The therefore required **bleeding time t7** has to be determined by trials at the customer plant. These trials are in responsibility of the customer and have to be executed by them self.

为了将标准排气过程之后仍然残留在轮缸制动回路中的空气排出，必须进行下列步骤：首先在制动踏板踩下之前锁紧前左轮的排气螺母，再用力踩下制动踏板以产生一个很高的制动系统压力，并随后打开下一个排气螺母。由

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于系统压力很高，残留空气将被压缩并在排气螺母打开后短时间内排出。残留空气连同制动液将依次通过ABS/ESP液压单元，制动管，轮缸，卡钳和排气螺母被排气设备排出。所需排气时间**t7**必须在客户工厂通过试验获得。这些试验应由客户自行完成。

The **handling time t70** depends on the operator and is required to connect the bleeding bottle respectively to open and close the bleeding screw.
操作时间**t70**取决于操作者连接排气头和打开锁紧排气螺母的时间。

The with x marked **timing values** depend on the size of the brake system, the adjusted bleeding pressure respectively the conditions at the vehicle (e.g. accessibility of the bleeding screws, etc). These times must be determined by trials executed at the customer assembly plant. The handling time must be determined together with the operators. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的时间值取决于制动系统的尺寸，排气压力和车辆的具体情况（如操作排气螺母的便捷性等）。该时间必须在客户装配工厂通过试验获得。操作时间也必须通过操作者试验确定。这些试验应由客户自行完成。

The with y marked **pressure value** is to be inferred from the chapter **Process data (pressure and time values)** and **Diagnostic data (timing values).**
数值为y的压力值可由“过程数据（压力和时间值）”和“诊断数据（时间值）”相关章节得出。

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3.8 Process step 8: Handling 2 过程步骤 8：操作 2

Handling 2 操作 2		
Operator: 操作者：		
☞ Release the vehicle 放下车辆 ☞ Switch off the bleeding device and disconnect it from the brake fluid reservoir 关闭排气设备并将其从储液壶上断开 ☞ Switch off the ignition 关闭点火开关 ☞ Disconnect the electrical connections (diagnostic connector) 断开电气连接（诊断接头） ☞ Correct the brake fluid level respectively fill it till the MAX sign is reached 加注制动液面至最高标记处 ☞ Mount the wheels of the vehicle 重新安好车轮		
Process step:	**Parameter:**	**Physical values:**
8	Handling time t8 操作时间 t8	x s

Notes: 注意：

The with x marked **handling time t8** depends on the installation position of the brake fluid reservoir, the position of the diagnosis interface OBD2 in the vehicle interior and the conditions at the vehicle. The handling time t8 must be determined together with the operators by trials. These trials are in responsibility of the customer and have to be executed by them self.
数值为x的**操作时间t8**取决于制动液储液壶安装位置、诊断接口OBD2在车内的位置以及车辆状况。同时，操作时间**t8**还必须经过操作者试验确认。这些试验应由客户自行完成。

At first the operator has to switch off the ignition before the diagnostic connector is disconnected.
在诊断接头断开之前操作者必须首先关闭点火开关。

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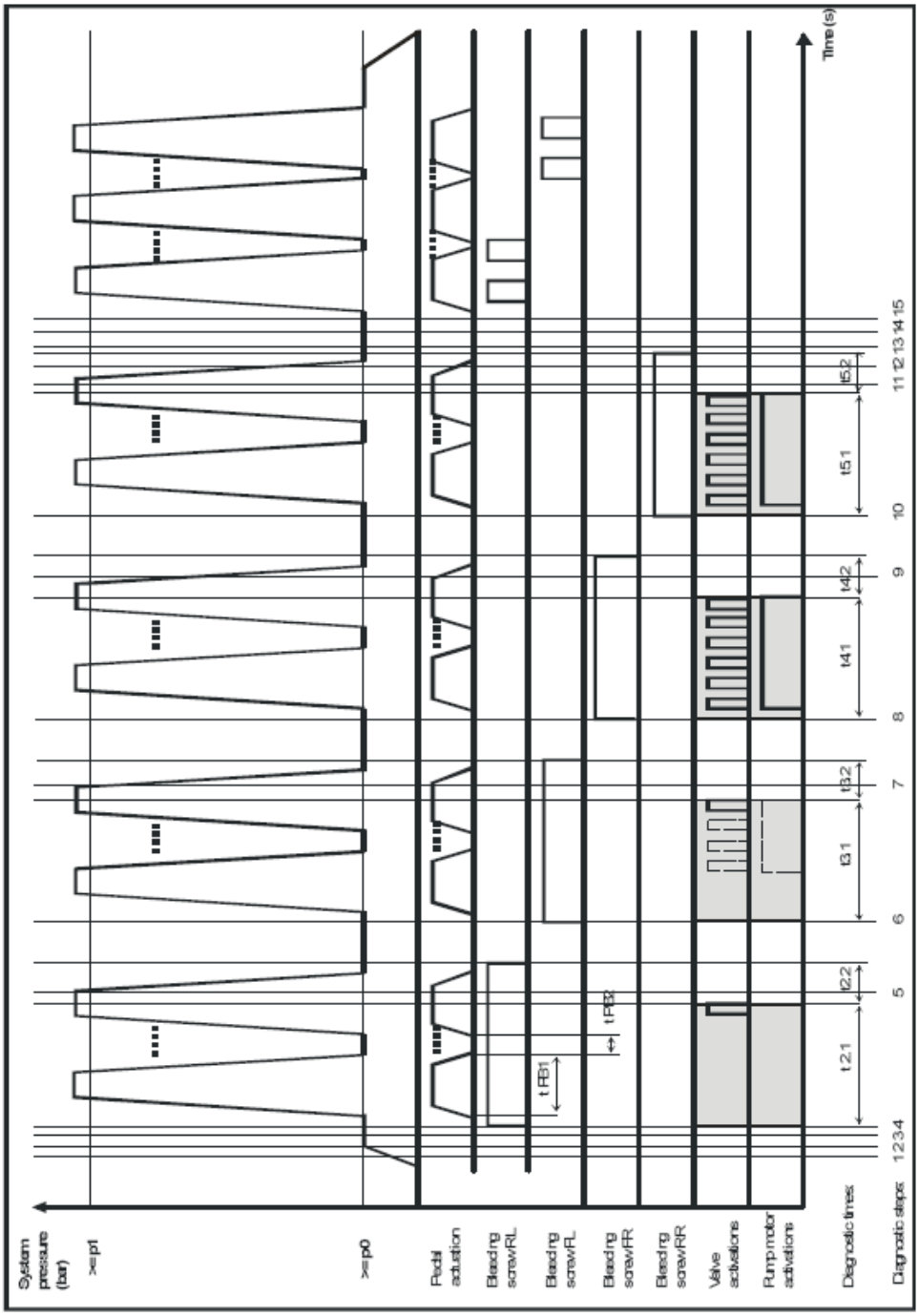
4 Description of all required diagnostic steps 所有必需的诊断步骤描述

Diagram of the sequence of all required diagnostic functions, valve and pump motor activations for G9
9代产品所有必需的诊断功能、阀动作和泵马达动作顺序示意图：



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Notes: 注意：

The bleeding screw of the respective wheel has to stay opened as long as the **wheel associated valve and pump motor activations** are finished. After that the respective wheel has to bleed a additional time as long as the brake fluid is free of air bubbles. The therefore required **waiting periods t2.2, t3.2, t4.2 and t5.2** depend on the size of the base brake system. These times have to be determined by trials with serial parts. These trials are in responsibility of the customer and have to be executed by them self.
各个车轮的排气螺母必须保持打开，直至该车轮的关联阀和泵马达的动作都结束，并额外保持一段时间以确保制动液中不再有气泡，之后排气螺母才能锁紧。各车轮对应的等待阶段时间**t2.2, t3.2, t4.2**和**t5.2**取决于基础制动系统的大小。这些时间必须由量产件试验来确定，试验应由客户自行完成。

The details to the valve and pump motor activations for the wheel
rear left (RL) = valve activations for the wheel RL = phase duration t2.1 + waiting period t2.2,
front left (FL) = valve activations (respectively pump motor activation for a 3 channel ABS) for the wheel FL = phase duration t3.1 + waiting period t3.2,
front right (FR) = valve and pump motor activations for the wheel FR = phase duration t4.1_ABS respectively t4.1_ESP + waiting period t4.2,
rear right (RR) = valve and pump motor activations for the wheel RR = phase duration t t5.1_ABS respectively t5.1_ESP + waiting period t5.2
can be taken from the chapter
Diagnosis step 4: Start of the valve activations for the wheel RL
Diagnosis step 6: Start of the valve activations (respectively pump motor activation at a 3 channel ABS) for the wheel FL
Diagnosis step 8: Start of the valve and pump motor activations for the wheel FR
Diagnosis step 10: Start of the valve and pump motor activations for the wheel RR.
各车轮的阀和泵马达具体动作：
后左轮（RL）= 后左轮阀动作时间 = 阶段持续时间t2.1 + 等待阶段时间t2.2，
前左轮（FL）= 前左轮阀动作时间（加上3通道ABS的泵马达动作时间） = 阶段持续时间t3.1 + 等待阶段时间t3.2，
前右轮（FR）= 前右轮阀和泵马达动作时间 = 阶段持续时间t4.1_ABS和t4.1_ESP + 等待阶段时间t4.2，
后右轮（RR）= 后右轮阀和泵马达动作时间 = 阶段持续时间t5.1_ABS和t5.1_ESP + 等待阶段时间t5.2
可参考下列章节：
诊断步骤4：开始后左轮阀动作
诊断步骤6：开始前左轮阀动作（加上3通道ABS的泵马达动作）
诊断步骤8：开始前右轮阀和泵马达动作
诊断步骤10：开始后右轮阀和泵马达动作

The wheel-associated valve and pump motor activations were defined by Bosch to ensure the best possible bleeding result with regard to the ABS / ESP hydraulic modulator. Deviations to the given sequence can lead to a deterioration of the entire bleeding result.
博世设定的车轮关联阀和泵马达的动作顺序可以确保ABS/ESP液压调节器的排气效果最佳。任何与指定顺序不符之处都可能导致排气效果不良。

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4.1 Diagnosis step 1: Start of communication 诊断步骤 1：开始通讯

Start of communication 开始通讯	
Diagnostic unit 诊断单元	ECU 电子控制单元
Start of communication 开始通讯	Change into the customer specific diagnostic mode 进入客户规范诊断模式
More information about the chapter "start of communication" is to be inferred from the customer specific documentations. 关于"开始通讯"更多信息请参考客户规范文件。	

Notes: 注意

The communication between the tester and ECU can be take place immediately after switching on of the power supply or ignition.
打开电源或点火开关后将立刻开始诊断设备与ECU之间的通讯。

In case of a not successful execution of the function „start of communication", it should be repeated minimum 3 times before the process is interrupted.
如果"开始通讯"功能执行不成功，在中断过程之前必须至少重复执行3次。

More details to the start of communication are to be inferred from the customer specific documentations.
关于"开始通讯"更多信息请参考客户规范文件。

Start of communication successful ⇒ Continue
Start of communication not successful ⇒ Abort, print out fault message
开始通讯成功 ⇒ 继续
开始通讯不成功 ⇒ 退出，打印错误信息

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4.2 Diagnosis step 2: ECU identification 诊断步骤 2：电子控制单元识别

ECU identification 电子控制单元识别	
Diagnostic unit 诊断单元	ECU 电子控制单元
ECU identification 电子控制单元识别	Output of the requested ECU data. 电子控制单元数据输出
More information about the chapter “ECU identification” is to be inferred from the customer specific documentations. 关于“电子控制单元识别”更多信息请参考客户规范文件。	

Notes: 注意：

The to be evaluated ECU identification data must be confirmed by the customer and its assembly plants.
待评估的电子控制单元识别数据必须由客户及其装配工厂确认。

The ECU identification data must be compared with the in the tester software stored desired ECU data. This is required to select the right valve activation sequence. The ECU identification data might be changed in case of an ABS/ESP software update respectively modification, therefore it must be easily possible with editor menu to update these data.
电子控制单元识别数据必须和诊断设备中存储的默认数据进行对比，以确保所选择的阀动作顺序正确无误。当 ABS/ESP 软件升级时，电子控制单元识别数据将发生变更，因此这些数据应当能用编辑菜单进行方便的升级。

More details to the chapter “ECU identification” are to be inferred from the customer specific documentations.
关于“电子控制单元识别”更多信息请参考客户规范文件。

Known system ⇒ Continue
Unknown system ⇒ Abort, print out fault message
已知系统 ⇒ 继续
未知系统 ⇒ 退出，打印错误信息

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4.3 Diagnosis step 3: Evaluation of the fault code memory 诊断步骤 3：故障码内存评估

Evaluation of the fault code memory 故障码内存评估	
Diagnostic unit 诊断单元	ECU 电子控制单元
Evaluation of all relevant fault code entries before the process is started. (all for the repair bleed process relevant fault codes can be taken from the table in the chapter Diagnosis step 12: Evaluation of the fault code memory. 在排气开始之前对所有相关故障码记录进行评估（所有与人工排气过程相关的故障码可参考诊断步骤 12：故障码内存评估）	Output of all stored fault codes 输出所有存储的故障码
More information about the chapter “Evaluation of the fault code memory” is to be inferred from the customer specific documentations. 关于“故障码内存评估”更多信息请参考客户规范文件。	

Notes: 注意：

First of all the fault code memory should be checked for relevant fault codes, because different faults cannot be detected by evaluation of the valve and pump motor status information. More details to the evaluation of the fault code memory can be taken from the chapter Diagnosis step 12: Evaluation of the fault code memory or the customer specific documentations.
首先应当检查故障码内存看是否存在人工排气相关故障码，因为只对阀和泵马达状态进行评估时有些故障无法发现。关于“故障码内存评估“更多信息请参考诊断步骤 12：故障码内存评估或客户规范文件。

None relevant fault code stored ⇒ Continue
Relevant fault code stored ⇒ Abort, print out fault message
相关故障码未存储 ⇒ 继续
相关故障码存储 ⇒ 退出，打印错误信息

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4.4 Diagnosis step 4: Start of the valve activations for the wheel RL 诊断步骤 4：开始后左轮阀动作

Start of the valve activations 开始阀动作	
Diagnostic unit 诊断单元	ECU 电子控制单元
Start of the valve activations for the wheel RL 开始后左轮阀动作	Activation of the valves 启动阀
More information about the chapter “Start of the valve activations for the wheel RL“ is to be inferred from the customer specific documentations. 关于“开始后左轮阀动作”更多信息请参考客户规范文件。	

Notes: 注意

The required valve activations for the wheel **RL** depend on the brake split configuration respectively the system (ABS/ESP). Details to the possible brake split configurations can be taken from the chapter Table of valve assignment depending on the brake circuit configuration.
后左轮阀动作取决于制动回路分布类型和整个ABS/ESP系统。关于制动回路分布类型更多信息请参考不同制动回路分布类型下的阀分配表。

After starting of the repair bleeding process with a **4-channel ABS / ESP system** the valve activations for the wheel RL will be started with activation of the **valve $AV_{P1.1}$** after a defined **waiting time t_{AV0}** for the **valve activation time t_{AV1}**. This is required to press brake fluid in a part of the secondary circuit of the ABS / ESP hydraulic modulator. After the specified valve activations are finished, RL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time t2.2** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t2** (= diagnosis time) for the wheel RL consists of the **phase duration t2.1** and **waiting time t2.2**.
在4通道ABS/ESP系统排气过程开始后，后左轮阀动作如下：在规定的等待时间**t_{AV0}**之后阀**$AV_{P1.1}$**启动**t_{AV1}**的时间，这是为了将制动液压入ABS/ESP液压单元的部分第二回路中。在指定的阀动作结束后，后左轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**t2.2**取决于车辆状况和制动系统设计，并需由客户规定。整个后左轮的排气时间**t2**（=诊断时间）包括阶段持续时间**t2.1**和等待时间**t2.2**两部分。

At a 3-channel ABS System no valve activations are executed **at the wheel RL, therefore the whole bleeding time t2** (= diagnosis time) only consists of the **waiting time t2.2**.
在3通道ABS系统中后左轮没有阀需要启动，因此整个排气时间**t2**（=诊断时间）只包括等待时间**t2.2**。

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Diagram for the wheel RL for a 4-channel ABS and ESP system:
4通道ABS/ESP系统后左轮排气示意图：

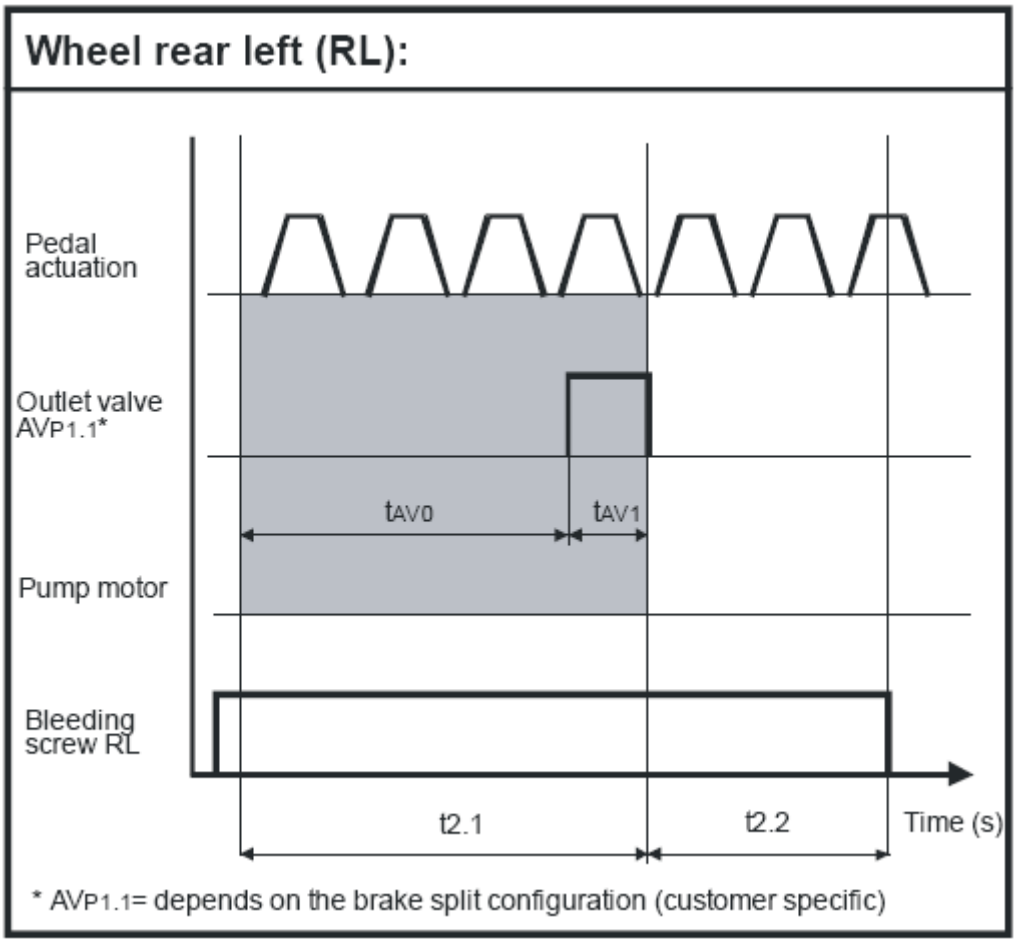

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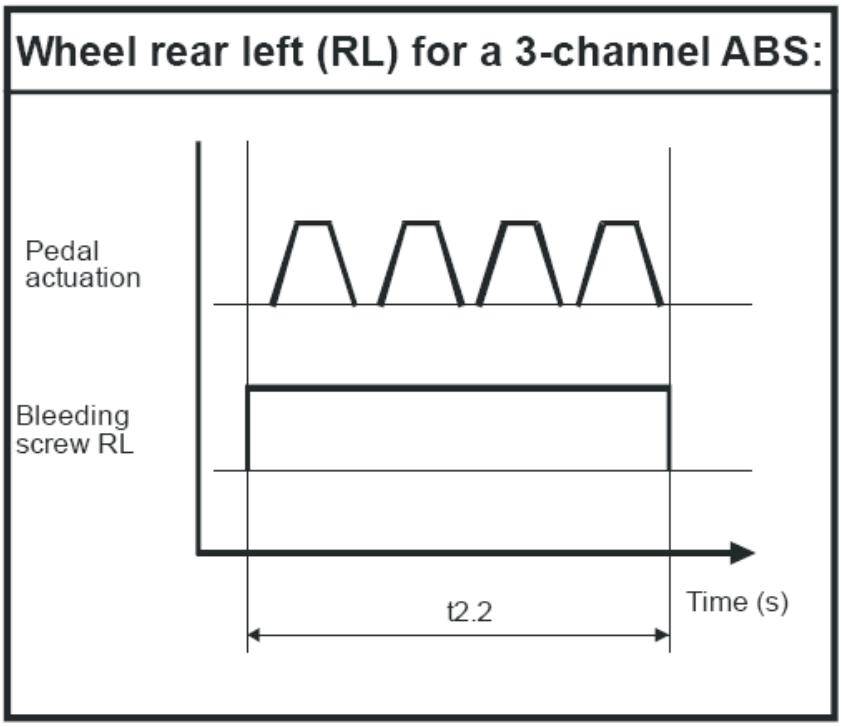
Diagram for the wheel RL for a 3-channel ABS system:
3通道ABS系统后左轮排气示意图：



Diagnostic parameter: 诊断参数：

Parameter:	Physical values:
Valve activation time t_{AV0} 阀动作时间t_{AV0}	y s
Valve activation time t_{AV1} 阀动作时间t_{AV1}	y s
Phase duration $t_{2.1}$ 阶段持续时间$t_{2.1}$	y s
Waiting time $t_{2.2}$ 等待时间$t_{2.2}$	x s

Notes: 注意：
The with x marked time value depends on the size of the base brake system and pressure p0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer.
数值为x的时间值取决于基础制动系统的尺寸和压力值p0。该时间必须在客户工厂通过试验获得，这些试验应由客户自行完成。

All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values).**
数值为y的时间值可由“诊断数据（时间值）”相关章节得出。

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4.5 Diagnosis step 5: Evaluation of the status of the valve activation 诊断步骤 5：阀动作状态评估

Evaluation of the status of the valve activation 阀动作状态评估	
Diagnostic unit 诊断单元	ECU 电子控制单元
Evaluation of the status of the valve activation 阀动作状态评估	Output of the status of the valve activation 输出阀动作状态
More information about the chapter “Evaluation of the status of the valve activation” is to be inferred from the customer specific documentations. 关于“阀动作状态评估”更多信息请参考客户规范文件。	

Notes: 注意：

To verify if the valve activation was successful executed and finished, the status of the valve activations have to be evaluated. The required diagnostic function is customer specific.
为了确认阀动作是否顺利执行完毕，必须评估阀动作状态。该诊断功能为客户指定。

More details to the chapter “Evaluation of the status of the valve activation” can be taken from the customer specific documentations.
关于“阀动作状态评估”更多信息请参考客户规范文件。

Following ECU responses (examples) are possible: 可能的ECU回复如下：

Valve activation is still running 阀动作进行中
Valve activation is successful finished 阀动作顺利完成
Valve activation interrupted 阀动作中断
Valve activation is unsuccessful finished (fault detected) 阀动作完成但不成功（检测到错误）

Important note to the interrogation of the system status (valve relay): 询问系统状态（阀继电器）注意事项：
With generation 9 the valve relay is always switched off in the normal state. In dependence of the selected diagnostic mode or first with starting of valve activation the valve relay is turned on. Therefore an interrogation of the valve relay to judge the system status is only limited helpful. However this evaluation can be applied to ensure a correct activation nevertheless in the individual or customer specific case, but this important remark must always be considered.

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9 代系统的阀继电器在正常状态下总是断开的。选择诊断模式或启动电磁阀时，阀继电器都将接通。因此询问阀继电器状态只能在一定程度上判断出系统状态。尽管这种判断方法只适用于个别情况或指定客户，但是在任何情况下都应予以重视。

Status of the valve activation correct ⇒ Continue
Status of the valve activation during the repair bleeding process incorrect ⇒ Abort, print out fault message

阀动作状态正确 ⇒ 继续
人工排气过程阀动作状态不正确 ⇒ 退出，打印错误信息

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4.6 Diagnosis step 6: Start of the valve activations (respectively pump motor activation at 3 channel ABS) for the wheel FL 诊断步骤 6：开始前左轮阀动作（加上 3 通道 ABS 的泵马达动作）

Start of the valve activations 开始阀动作	
Diagnostic unit 诊断单元	ECU 电子控制单元
Start of the valve activations for the wheel FL 开始前左轮阀动作	Activation of the valves 启动阀
More information about the chapter “Start of the valve activations (respectively pump motor activation at a 3 channel ABS) for the wheel FL“ is to be inferred from the customer specific documentations. 关于“开始前左轮阀动作（加上3通道ABS的泵马达动作）”更多信息请参考客户规范文件。	

Notes: 注意：

The required valve activations for the wheel **FL** depend on the brake split configuration respectively the system (ABS/ESP). Details to the possible brake split configurations can be taken from the chapter Table of valve assignment depending on the brake circuit configuration.
前左轮阀动作取决于制动回路分布类型和整个ABS/ESP系统。关于制动回路分布类型更多信息请参考不同制动回路分布类型下的阀分配表。

After starting of the repair bleeding process with a **4-channel ABS / ESP system** the valve activations for the wheel FL will be started with activation of the **valve AV$_{P2.1}$** after a defined **waiting time t$_{AV0}$** for the **valve activation time t$_{AV1}$**. This is required to press brake fluid in a part of the secondary circuit of the ABS / ESP hydraulic modulator. After the specified valve activations are finished, FL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time t3.2** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t3** (= diagnosis time) for the wheel FL consists of the **phase duration t3.1** and **waiting time t3.2**.
在4通道ABS/ESP系统排气过程开始后，前左轮阀动作如下：在规定的等待时间**t$_{AV0}$**之后阀**AV$_{P2.1}$**启动**t$_{AV1}$**的时间，这是为了将制动液压入ABS/ESP液压单元的部分第二回路中。在指定的阀动作结束后，前左轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**t3.2**取决于车辆状况和制动系统设计，并需由客户规定。整个前左轮的排气时间**t3**（=诊断时间）包括阶段持续时间**t3.1**和等待时间**t3.2**两部分。

After starting of the repair bleeding process with a **3-channel ABS system** the valve activations for the wheel FL will be started with activation of the valves **EV$_{P2.1}$**, **EV$_{P2.2}$** and **AV$_{P2.1}$** and the **pump motor** after a defined **waiting time t$_{AV0}$**. The pump motor and the valves EV$_{P2.1}$ and EV$_{P2.2}$ are activated for the **pump motor activation time t$_{PM1.1}$** (= valve activation time). The activation of the pump motor is required to supply the pump element at the rear axle circuit with brake fluid. To the same time the valve AV$_{P2.1}$ is alternate activated on and off for the **valve activation time t$_{AVon}$** and **valve activation time t$_{AVoff}$**. After that the valve **AV$_{P2.2}$** is activated for the **valve activation time t$_{AV1}$**. After the specified valve activations are finished, FL must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time**

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t3.2 depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t3** (= diagnosis time) for the wheel FL consists of the **phase duration t3.1_3K_ABS** and **waiting time 3.2**.

在3通道ABS系统排气过程开始后，前左轮阀动作如下：在规定的等待时间t_{AV0}之后阀$EV_{P2.1}$，$EV_{P2.2}$ 和$AV_{P2.1}$以及泵马达启动，其中阀$EV_{P2.1}$，$EV_{P2.2}$ 和泵马达启动$t_{PM1.1}$的时间（=阀启动时间）。启动泵马达是为了给后轴回路的泵单元提供制动液。在此期间，阀$AV_{P2.1}$交替启动（每次启动时间t_{AVon}）与关闭（每次关闭时间t_{AVoff}）。随后阀$AV_{P2.2}$将启动t_{AV1}的时间。在指定的阀动作结束后，前左轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**t3.2**取决于车辆状况和制动系统设计，并需由客户规定。整个前左轮的排气时间**t3**（=诊断时间）包括阶段持续时间**t3.1_3K_ABS**和等待时间**t3.2**两部分。

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Diagram for the wheel FL for a 4-channel ABS and ESP system:
4通道ABS/ESP系统前左轮排气示意图：

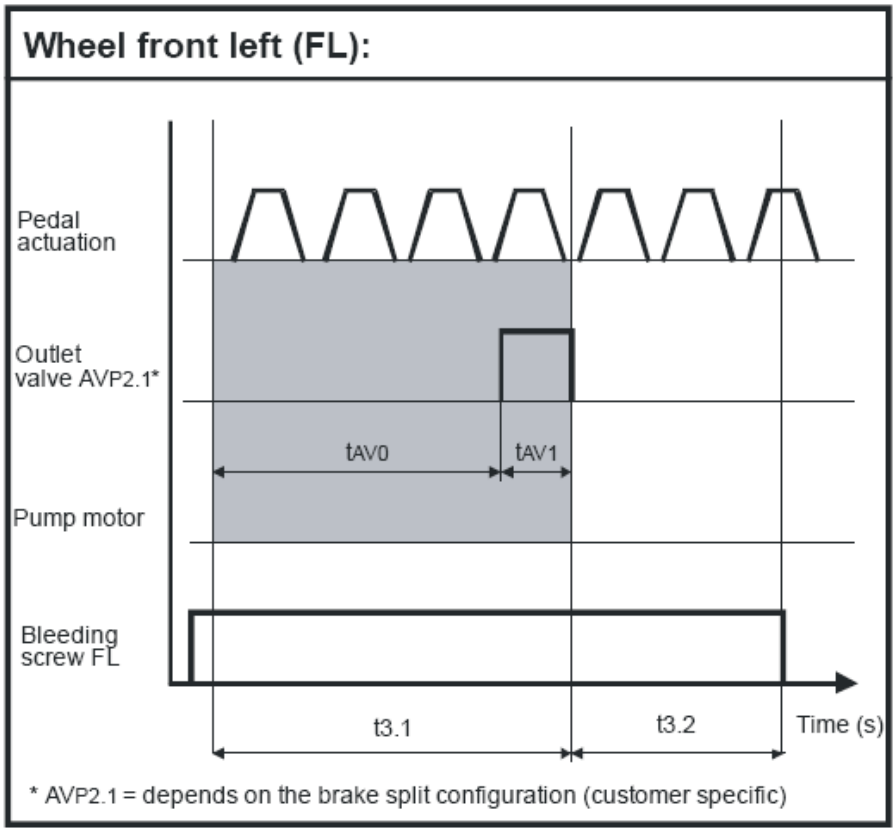

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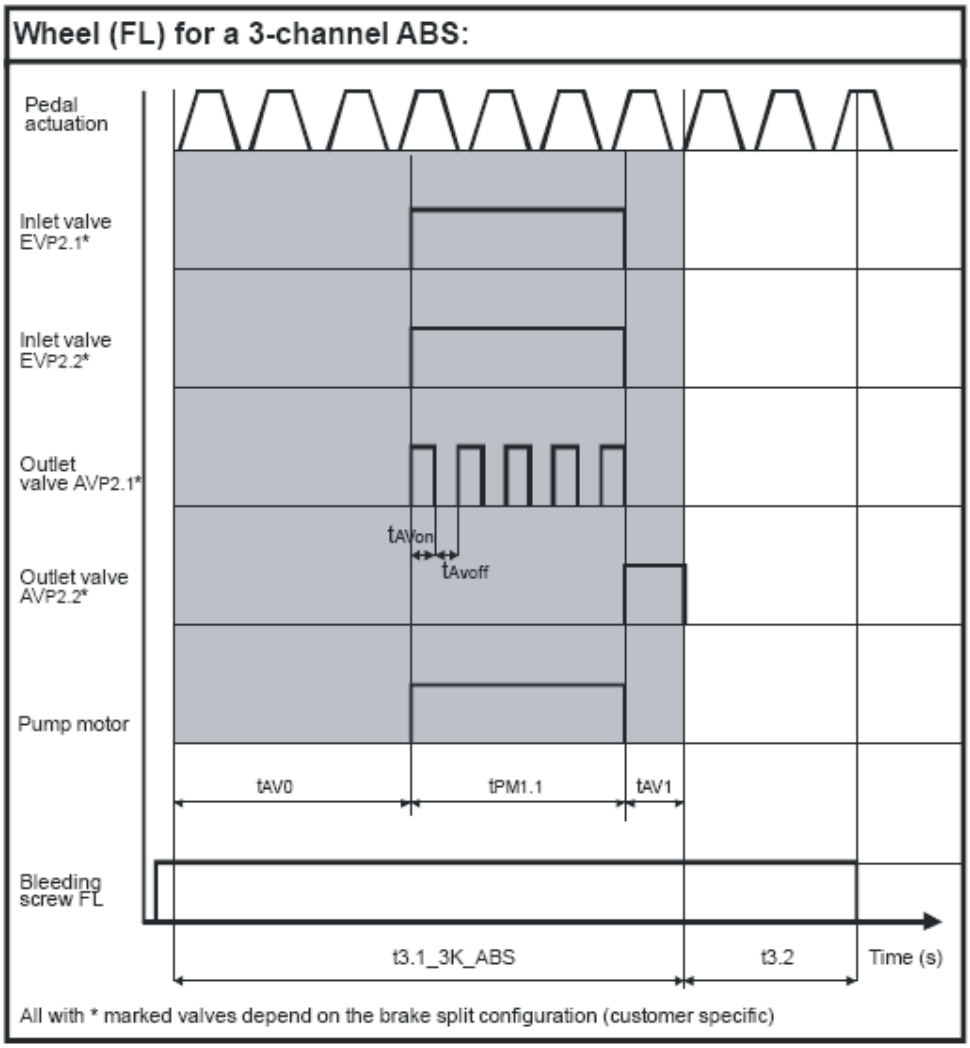
Diagram for the wheel FL for a 3-channel ABS system:
3通道ABS系统前左轮排气示意图：



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Diagnostic parameter: 诊断参数：

Parameter:	Physical values:
Valve activation time t_{AV0} 阀动作时间t_{AV0}	y s
Valve activation time t_{AV1} 阀动作时间t_{AV1}	y s
Valve activation time t_{AVon} 阀动作时间t_{AVon}	y s
Valve activation time t_{AVoff} 阀动作时间t_{AVoff}	y s
Pump motor activation time $t_{PM1.1}$ 泵马达动作时间$t_{PM1.1}$	y s
Phase duration $t_{3.1}$ 阶段持续时间$t_{3.1}$	y s
Phase duration $t_{3.1_3K_ABS}$ 阶段持续时间$t_{3.1_3K_ABS}$	y s
Waiting time $t_{3.2}$ 等待时间$t_{3.2}$	x s

Notes: The with x marked time value depends on the size of the base brake system and pressure p0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer. All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values)**.
注意：数值为x的时间值取决于基础制动系统的尺寸和压力值p0。该时间必须在客户工厂通过试验获得，这些试验应由客户自行完成。数值为y的时间值可由“诊断数据（时间值）”相关章节得出。

4.7 Diagnosis step 7: Evaluation of the status of the valve activation 诊断步骤 7：阀动作状态评估

Details can be taken from the chapter 4.5. 具体请参考4.5小节。

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4.8 Diagnosis step 8: Start of the valve and pump motor activations for the wheel FR
诊断步骤 8：开始前右轮阀和泵马达动作

Start of the valve activations 开始阀动作	
Diagnostic unit 诊断单元	ECU 电子控制单元
Start of the valve activations or the wheel FR 开始前右轮阀动作	Activation of the valves 启动阀
More information about the chapter “Start of the valve and pump motor activations for the wheel FR“ is to be inferred from the customer specific documentations. 关于“开始前右轮阀和泵马达动作”更多信息请参考客户规范文件。	

Notes: 注意：

The required valve activations for the wheel **FR** depend on the brake split configuration respectively the system (ABS/ESP). Details to the possible brake split configurations can be taken from the chapter Table of valve assignment depending on the brake circuit configuration.
前右轮阀动作取决于制动回路分布类型和整个ABS/ESP系统。关于制动回路分布类型更多信息请参考不同制动回路分布类型下的阀分配表。

Description for a 3-channel ABS system: 3通道ABS系统描述

After starting of the repair bleeding process the valve activations for the wheel FR will be started with activation of the valves **$EV_{P3.1}$**, **$EV_{P3.2}$** and **$AV_{P3.1}$** after a defined **waiting time t_{Delay1}** for the **valve activation time t_{AV2}**. After a further waiting time (= valve activation time **t_{AV2}** – pump motor activation time **$t_{PM1.1}$**) the pump motor is activated for the **pump motor activation time $t_{PM1.1}$**.The activation of the pump motor is required to supply the pump element at the front axle circuit with brake fluid. Immediately after all above described activations are finished the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop: After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P3.2}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P3.1}$** is activated with the same time (valve activation time t_{AV1}). After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers. Afterwards the specified valve activations are finished, FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_4** (= diagnosis time) for the wheel FR consists of the **phase duration $t_{4.1_ABS}$** and **waiting time $t_{4.2}$**.
在排气过程开始后，前右轮阀动作如下：在规定的等待时间**t_{Delay1}**之后，阀**$EV_{P3.1}$**，**$EV_{P3.2}$** 和**$AV_{P3.1}$**启动**t_{AV2}**的时间。再经过一段等待时间（=阀动作时间**t_{AV2}** – 泵动作时间**$t_{PM1.1}$**），泵马达将启动**$t_{PM1.1}$**的时

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间。启动泵马达是为了给前轴回路的泵单元提供制动液。在以上动作全部结束后，立即开始的第二回路排气将进行重复规定次数（≤ n1）的动作循环。每个循环包括下列阀动作：在规定等待时间t_{Delay1}之后，泵马达启动$t_{PM1.2}$的时间。同时，在等待时间t_{Delay1}结束后又经过t_{Delay2}的时间后，阀$AV_{P3.2}$将启动t_{AV1}的时间。在t_{AV1}的时间结束后，阀$AV_{P3.1}$也将启动相同时间（t_{AV1}）。在规定循环次数完成且所有阀动作结束之后，再经过t_{Delay1}的等待时间，泵马达将启动$t_{PM1.4}$的时间以排空蓄能仓。在指定的阀动作结束后，前右轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间$t_{4.2}$取决于车辆状况和制动系统设计，并需由客户规定。整个前右轮的排气时间t_4（=诊断时间）包括阶段持续时间$t_{4.1_ABS}$和等待时间$t_{4.2}$两部分。

Description for a 4-channel ABS system: 4通道ABS系统描述

After starting of the repair bleeding process the valve activations for the wheel FR will be started with activation of the valves **$EV_{P3.1}$**, **$EV_{P3.2}$**, **$AV_{P3.1}$** and **$AV_{P3.2}$** after a defined **waiting time t_{Delay1}** for the **valve activation time t_{AV2}**. After a further waiting time (= valve activation time t_{AV2} – pump motor activation time $t_{PM1.1}$) the pump motor is activated for the **pump motor activation time $t_{PM1.1}$**.The activation of the pump motor is required to supply the pump elements with brake fluid.
Immediately after all above described activations are finished the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop: After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $t_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P3.3}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P3.1}$** is activated with the same time (valve activation time t_{AV1}). After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers. Afterwards the specified valve activations are finished, FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_4** (= diagnosis time) for the wheel FR consists of the **phase duration $t_{4.1_ABS}$** and **waiting time $t_{4.2}$**.
在排气过程开始后，前右轮阀动作如下：在规定的等待时间t_{Delay1}之后，阀$EV_{P3.1}$，$EV_{P3.2}$，$AV_{P3.1}$和$AV_{P3.2}$启动t_{AV2}的时间。再经过一段等待时间（=阀动作时间t_{AV2} – 泵动作时间$t_{PM1.1}$），泵马达将启动$t_{PM1.1}$的时间。启动泵马达是为了给前轴回路的泵单元提供制动液。在以上动作全部结束后，立即开始的第二回路排气将进行重复规定次数（≤ n1）的动作循环。每个循环包括下列阀动作：在规定等待时间t_{Delay1}之后，泵马达启动$t_{PM1.2}$的时间。同时，在等待时间t_{Delay1}结束后又经过t_{Delay2}的时间后，阀$AV_{P3.3}$将启动t_{AV1}的时间。在t_{AV1}的时间结束后，阀$AV_{P3.1}$也将启动相同时间（t_{AV1}）。在规定循环次数完成且所有阀动作结束之后，再经过t_{Delay1}的等待时间，泵马达将启动$t_{PM1.4}$的时间以排空蓄能仓。在指定的阀动作结束后，前右轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间$t_{4.2}$取决于车辆状况和制动系统设计，并需由客户规定。整个前右轮的排气时间t_4（=诊断时间）包括阶段持续时间$t_{4.1_ABS}$和等待时间$t_{4.2}$两部分。

Description for an ESP system: ESP系统描述

After starting of the repair bleeding process the valve activations for the wheel FR will be started with activation of the valves **$EV_{P3.1}$** and **$EV_{P3.2}$** after a defined **waiting time t_{Delay1}** for the **valve activation time t_{AV3}**. In parallel the valves **$AV_{P3.1}$** and **$AV_{P3.2}$** are activated **t_{Delay1}** for the **valve**

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activation time t_{AV1}. Directly after the valve activation time t_{AV1} is expired the high pressure switching valves **$HSV_{P3.1}$** and **$HSV_{P3.2}$** are alternate activated with on/ off **(t_{HSVon} / t_{HSVoff})** for the **valve activation time $t_{HSV1.1}$**. In addition the pump motor is activated for the **pump motor activation time $t_{PM1.1}$**.The activation of the pump motor is required to supply the pump elements with brake fluid. Immediately after all above described activations are finished the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop: After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P3.3}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P3.1}$** is activated with the same time (valve activation time t_{AV1}). After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired the high pressure switching valve **$HSV_{P3.1}$** is alternate activated with on/ off **(t_{HSVon} / t_{HSVoff})** for the **valve activation time $t_{HSV1.2}$**. In parallel the pump motor is activated for the **pump motor activation time $t_{PM1.3}$** to bleed the respective priming circuit. In fact of that the pump motor activation time $t_{PM1.3}$ is longer than the valve activation time $t_{HSV1.2}$ all valves are closed at the end, therefore it is guaranteed that the accumulators will be discharged. Afterwards the specified valve activations are finished, FR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_4** (= diagnosis time) for the wheel FR consists of the **phase duration $t_{4.1_ESP}$** and **waiting time $t_{4.2}$**.

在排气过程开始后，前右轮阀动作如下：在规定的等待时间**t_{Delay1}**之后，阀**$EV_{P3.1}$**和**$EV_{P3.2}$**启动**t_{AV3}**的时间；同时阀**$AV_{P3.1}$**和**$AV_{P3.2}$**在等待**t_{Delay1}**后启动**t_{AV1}**的时间。在**t_{AV1}**时间结束后，高压开关阀**$HSV_{P3.1}$** 和 **$HSV_{P3.2}$**将在**$t_{HSV1.1}$**的时间内交替启动/关闭（启动/关闭时间：**t_{HSVon} / t_{HSVoff}**）；与此同时泵马达也启动了**$t_{PM1.1}$**的时间。启动泵马达是为了给泵单元提供制动液。在以上动作全部结束后，立即开始的第二回路排气将进行重复规定次数（≤ n1）的动作循环。每个循环包括下列阀动作：在规定等待时间**t_{Delay1}**之后，泵马达启动**$t_{PM1.2}$**的时间。同时，在等待时间**t_{Delay1}**结束后又经过**t_{Delay2}**的时间后，阀**$AV_{P3.3}$**将启动**t_{AV1}**的时间。在**t_{AV1}**的时间结束后，阀**$AV_{P3.1}$**也将启动相同时间（**t_{AV1}**）。在规定循环次数完成且所有阀动作结束之后，再经过**t_{Delay1}**的等待时间，高压开关阀**$HSV_{P3.1}$**将在**$t_{HSV1.2}$**的时间内交替启动/关闭（启动/关闭时间：**t_{HSVon} / t_{HSVoff}**），同时泵马达将启动**$t_{PM1.3}$**的时间以排空第一回路。因为泵动作时间**$t_{PM1.3}$**比阀动作时间**$t_{HSV1.2}$**长，所有阀在最后都将关闭，因此可以确保蓄能仓可以被排空。在指定的阀动作结束后，前右轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**$t_{4.2}$**取决于车辆状况和制动系统设计，并需由客户规定。整个前右轮的排气时间**t_4**（=诊断时间）包括阶段持续时间**$t_{4.1_ESP}$**和等待时间**$t_{4.2}$**两部分。

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Diagram for the wheel FR for 3-channel ABS system:
3通道ABS系统前右轮排气示意图：

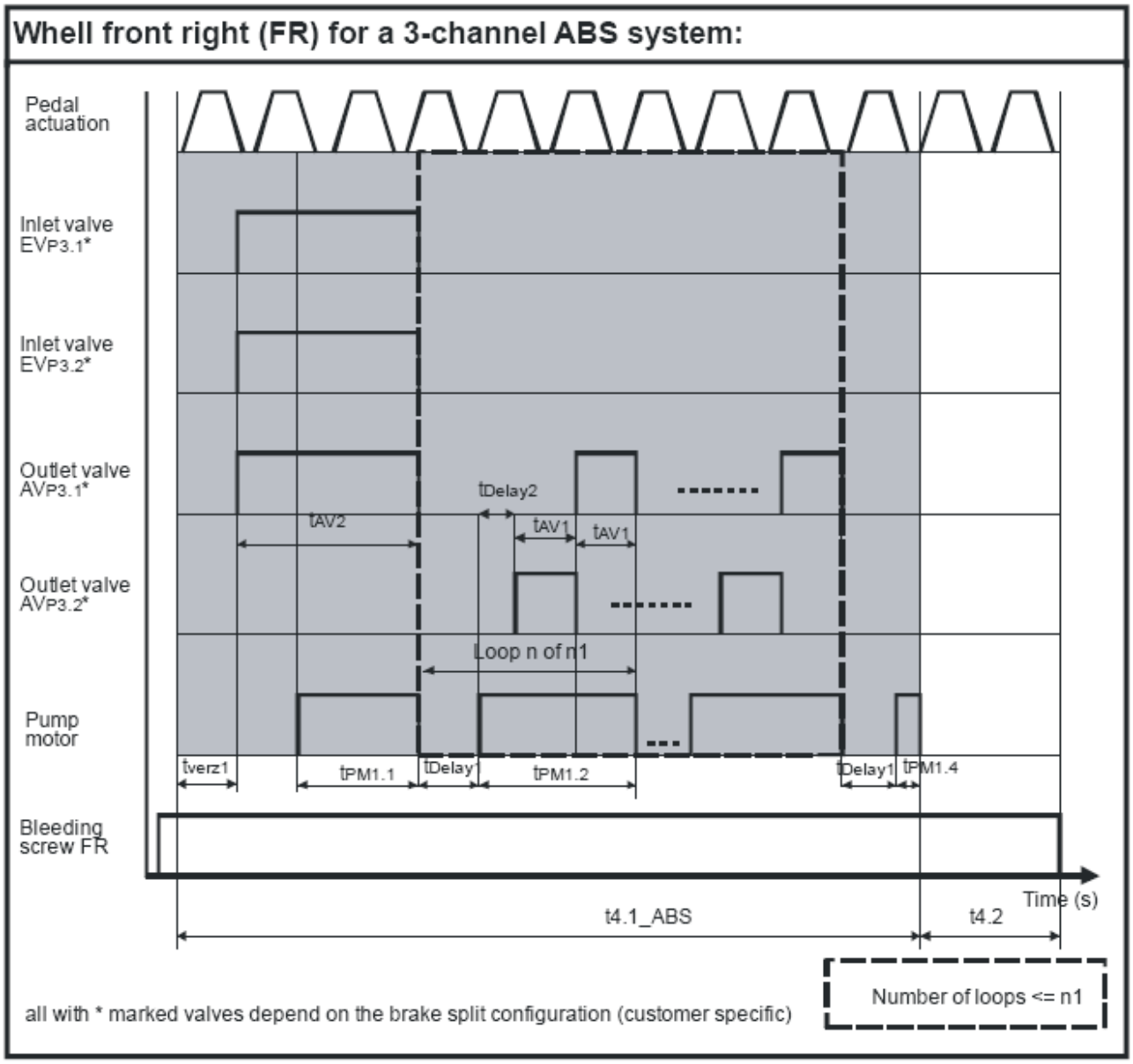

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Diagram for the wheel FR for 4-channel ABS system:
4通道ABS系统前右轮排气示意图：

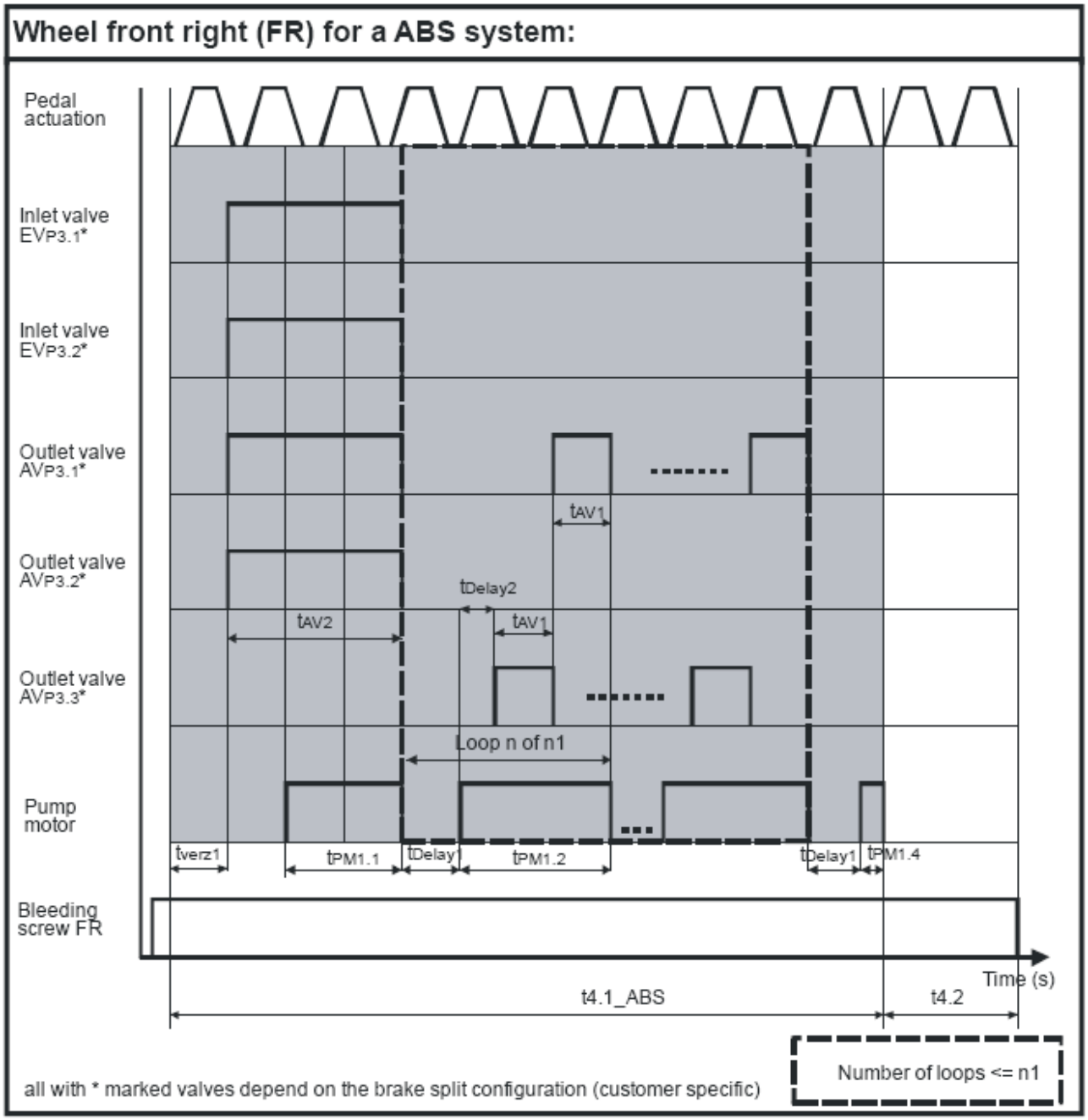

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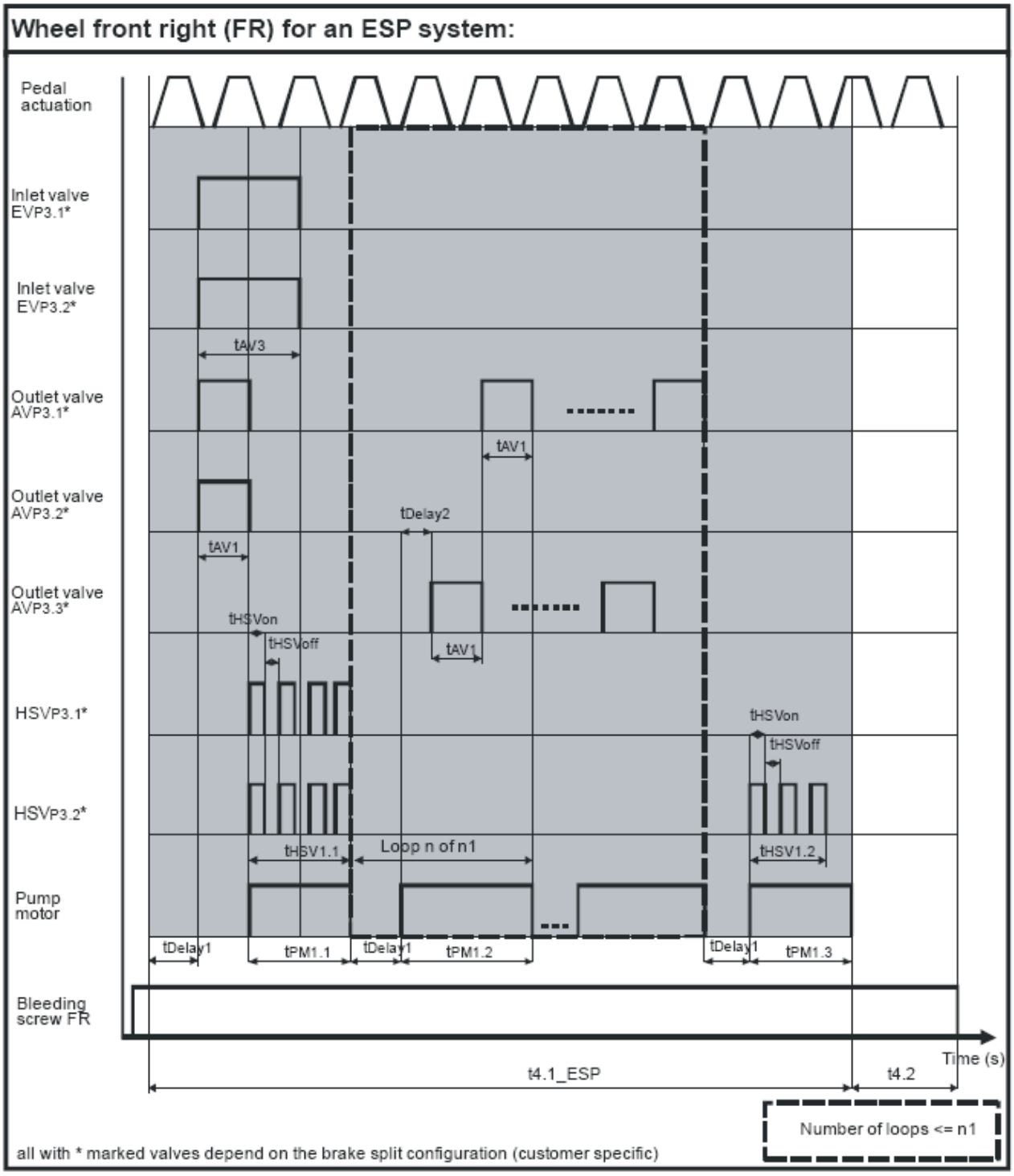
Diagram for wheel FR for a ESP system: ESP系统前右轮排气示意图：



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Diagnostic parameter: 诊断参数：

Parameter:	Physical values:
Valve activation time t_{AV1} 阀动作时间t_{AV1}	y s
Valve activation time t_{AV2} 阀动作时间t_{AV2}	y s
Valve activation time t_{AV3} 阀动作时间t_{AV3}	y s
Valve activation time $t_{HSV1.1}$ 阀动作时间$t_{HSV1.1}$	y s
Valve activation time $t_{HSV1.2}$ 阀动作时间$t_{HSV1.2}$	y s
Valve activation time t_{HSVon} 阀动作时间t_{HSVon}	y s
Valve activation time t_{HSVoff} 阀动作时间t_{HSVoff}	y s
Waiting time t_{Delay1} 等待时间t_{Delay1}	y s
Waiting time t_{Delay2} 等待时间t_{Delay2}	y s
Pump motor activation time $t_{PM1.1}$ 泵马达动作时间$t_{PM1.1}$	y s
Pump motor activation time $t_{PM1.2}$	y s
Pump motor activation time $t_{PM1.3}$	y s
Pump motor activation time $t_{PM1.4}$	y s
Number of loops n_1 循环次数n_1	y
Max. Phase duration $t_{4.1_ABS}$ 最长阶段持续时间$t_{4.1_ABS}$	y s
Max. Phase duration $t_{4.1_ESP}$ 最长阶段持续时间$t_{4.1_ESP}$	y s
Waiting time $t_{4.2}$ 等待时间$t_{4.2}$	x s

Notes: 注意：

The with x marked time value depends on the size of the base brake system and pressure p0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer. All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values).**
数值为x的时间值取决于基础制动系统的尺寸和压力值p0。该时间必须在客户工厂通过试验获得，这些试验应由客户自行完成。数值为y的时间值可由“诊断数据（时间值）”相关章节得出。

4.9 Diagnosis step 9: Evaluation of the status of the valve activation 诊断步骤 9：阀动作状态评估

Details can be taken from the chapter 4.5. 具体请参考4.5小节。

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4.10 Diagnosis step 10: Start of the valve and pump motor activations for the wheel RR 诊断步骤 10：开始后右轮阀和泵马达动作

Start of the valve activations 开始阀动作	
Diagnostic unit 诊断单元	ECU 电子控制单元
Start of the valve activations or the wheel RR 开始后右轮阀动作	Activation of the valves 启动阀
More information about the chapter “Start of the valve and pump motor activations for the wheel RR“ is to be inferred from the customer specific documentations. 关于“开始后右轮阀和泵马达动作”更多信息请参考客户规范文件。	

Notes: 注意：

Description for a 3-channel ABS system: 3通道ABS系统描述

After starting of the repair bleeding process for the wheel RR the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop: After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $_{PM1.5}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P4.1}$** is switched on for the **valve activation time t_{AV1}**. After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers. Afterwards the specified valve activations are finished, RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{5.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_5** (= diagnosis time) for the wheel RR consists of the **phase duration $t_{5.1_ABS}$** and **waiting time $t_{5.2}$**.
在后右轮排气过程中，对第二回路排气是通过重复规定次数（≤ n1）的动作循环来实现的。每个循环包括下列阀动作：在规定等待时间**t_{Delay1}**之后，泵马达启动**$t_{PM1.5}$**的时间。同时，在等待时间**t_{Delay1}**结束后又经过**t_{Delay2}**的时间后，阀**$AV_{P4.1}$**将启动**t_{AV1}**的时间。在规定循环次数完成且所有阀动作结束之后，再经过**t_{Delay1}**的等待时间，泵马达将启动**$t_{PM1.4}$**的时间以排空蓄能仓。在指定的阀动作结束后，后右轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**$t_{5.2}$**取决于车辆状况和制动系统设计，并需由客户规定。整个后右轮的排气时间**t_5**（=诊断时间）包括阶段持续时间**$t_{5.1_ABS}$**和等待时间**$t_{5.2}$**两部分。

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Description for a 4-channel ABS system: 4通道ABS系统描述

After starting of the repair bleeding process for the wheel RR the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop: After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P4.2}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P4.1}$** is activated with the same time (valve activation time t_{AV1}). After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired, the pump motor is activated for the **pump motor activation time $t_{PM1.4}$** to discharge the accumulator chambers. Afterwards the specified valve activations are finished, RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{5.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_5** (= diagnosis time) for the wheel RR consists of the **phase duration $t_{5.1_ABS}$** and **waiting time $t_{5.2}$**.

在后右轮排气过程中，对第二回路排气是通过重复规定次数（≤ n1）的动作循环来实现的。每个循环包括下列阀动作：在规定等待时间**t_{Delay1}**之后，泵马达启动**$t_{PM1.2}$**的时间。同时，在等待时间**t_{Delay1}**结束后又经过**t_{Delay2}**的时间后，阀**$AV_{P4.2}$**将启动**t_{AV1}**的时间。在**t_{AV1}**的时间结束后，阀**$AV_{P4.1}$**也将启动相同时间（**t_{AV1}**）。在规定循环次数完成且所有阀动作结束之后，再经过**t_{Delay1}**的等待时间，泵马达将启动**$t_{PM1.4}$**的时间以排空蓄能仓。在指定的阀动作结束后，后右轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**$t_{5.2}$**取决于车辆状况和制动系统设计，并需由客户规定。整个后右轮的排气时间**t_5**（=诊断时间）包括阶段持续时间**$t_{5.1_ABS}$**和等待时间**$t_{5.2}$**两部分。

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Description for an ESP system: **ESP系统描述**

After starting of the repair bleeding process for the wheel RR the bleeding of the respective secondary circuit is started with execution of the valve activations which are included in a loop. This loop is repeated for a defined number (number ≤ n1). Following valve activations are included in this loop: After a defined **waiting time t_{Delay1}** the pump motor is activated for the **pump motor activation time $_{PM1.2}$**. Directly afterwards the **waiting times** t_{Delay1} and **t_{Delay2}** are expired the valve **$AV_{P4.2}$** is switched on for the **valve activation time t_{AV1}**. After the valve activation time t_{AV1} is expired the valve **$AV_{P4.1}$** is activated with the same time (valve activation time t_{AV1}). After the selected number of loops respectively all valve activations are finished and a further **waiting time t_{Delay1}** is expired the high pressure switching valve **$HSV_{P4.1}$** is alternate activated with on/ off **(t_{HSVon} / t_{HSVoff})** for the **valve activation time $t_{HSV1.2}$**. In parallel the pump motor is activated for the **pump motor activation time $t_{PM1.3}$** to bleed the respective priming circuit. In fact of that the pump motor activation time $t_{PM1.3}$ is longer than the valve activation time $t_{HSV1.2}$ all valves are closed at the end, therefore it is guaranteed that the accumulators will be discharged. Afterwards the specified valve activations are finished, RR must be bleed as long as the brake fluid is seen free of air bubbles at the bleeding screw. This additional **waiting time $t_{4.2}$** depends on the vehicle and the design of the brake system. This time has to be specified by the customer. The whole **bleeding time t_5** (= diagnosis time) for the wheel RR consists of the **phase duration $t_{5.1_ESP}$** and **waiting time $t_{5.2}$**.

在后右轮排气过程中，对第二回路排气是通过重复规定次数（≤ n1）的动作循环来实现的。每个循环包括下列阀动作：在规定等待时间**t_{Delay1}**之后，泵马达启动**$t_{PM1.2}$**的时间。同时，在等待时间**t_{Delay1}**结束后又经过**t_{Delay2}**的时间后，阀**$AV_{P4.2}$**将启动**t_{AV1}**的时间。在**t_{AV1}**的时间结束后，阀**$AV_{P4.1}$**也将启动相同时间（**t_{AV1}**）。在规定循环次数完成且所有阀动作结束之后，再经过**t_{Delay1}**的等待时间，高压开关阀**$HSV_{P4.1}$**将在**$t_{HSV1.2}$**的时间内交替启动/关闭（启动/关闭时间：**t_{HSVon} / t_{HSVoff}**），同时泵马达将启动**$t_{PM1.3}$**的时间以排空第一回路。因为泵动作时间**$t_{PM1.3}$**比阀动作时间**$t_{HSV1.2}$**长，所有阀在最后都将关闭，因此可以确保蓄能仓可以被排空。在指定的阀动作结束后，后右轮必须接着排气直至排气螺母处流出的制动液没有气泡为止。这一额外的等待时间**$t_{4.2}$**取决于车辆状况和制动系统设计，并需由客户规定。整个后右轮的排气时间**t_5**（=诊断时间）包括阶段持续时间**$t_{5.1_ESP}$**和等待时间**$t_{5.2}$**两部分。

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Diagram for wheel RR for a 3-channel ABS system:
3通道ABS系统后右轮排气示意图：

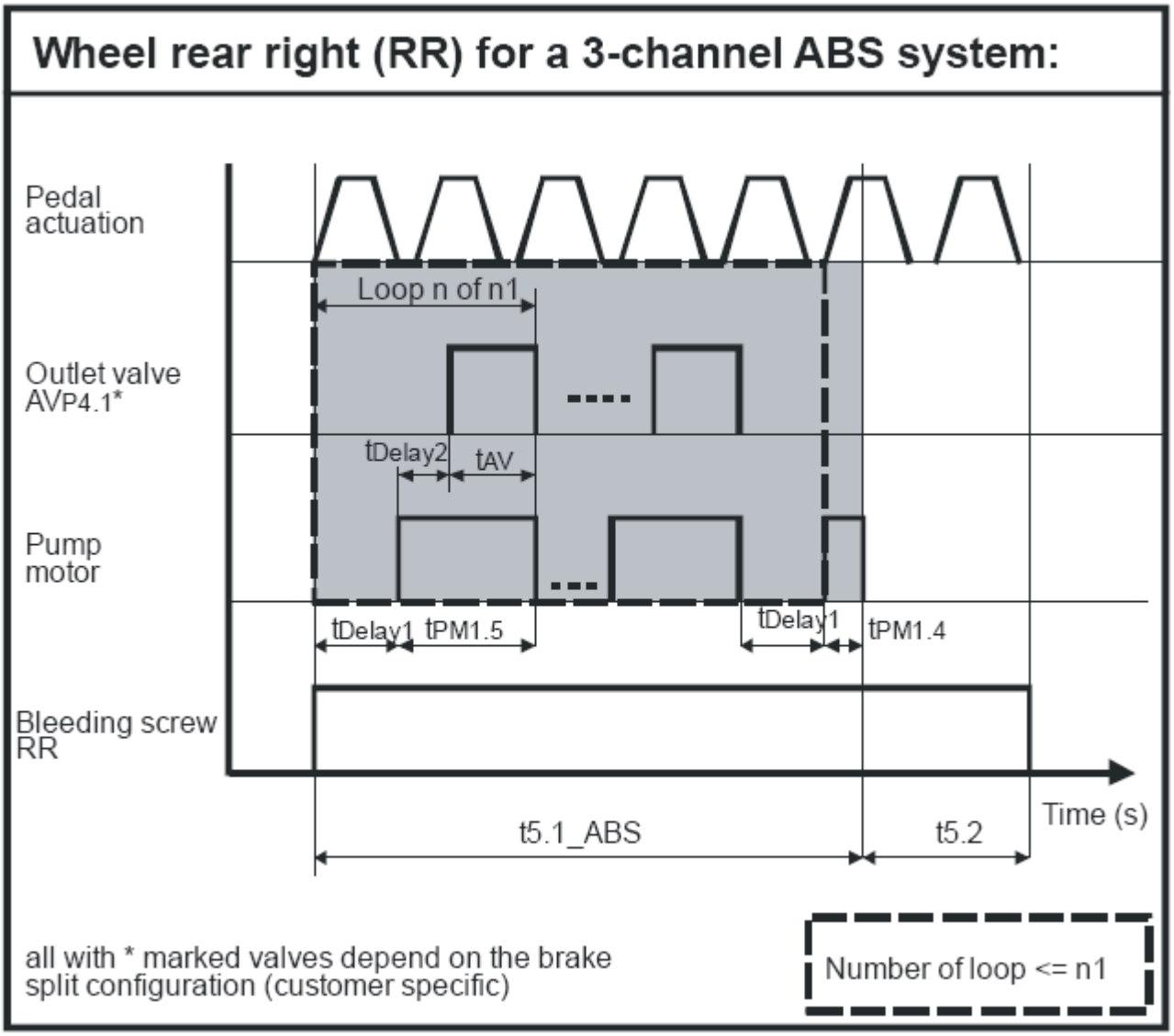

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Diagram for wheel RR for a 4-channel ABS system:
4通道ABS系统后右轮排气示意图：

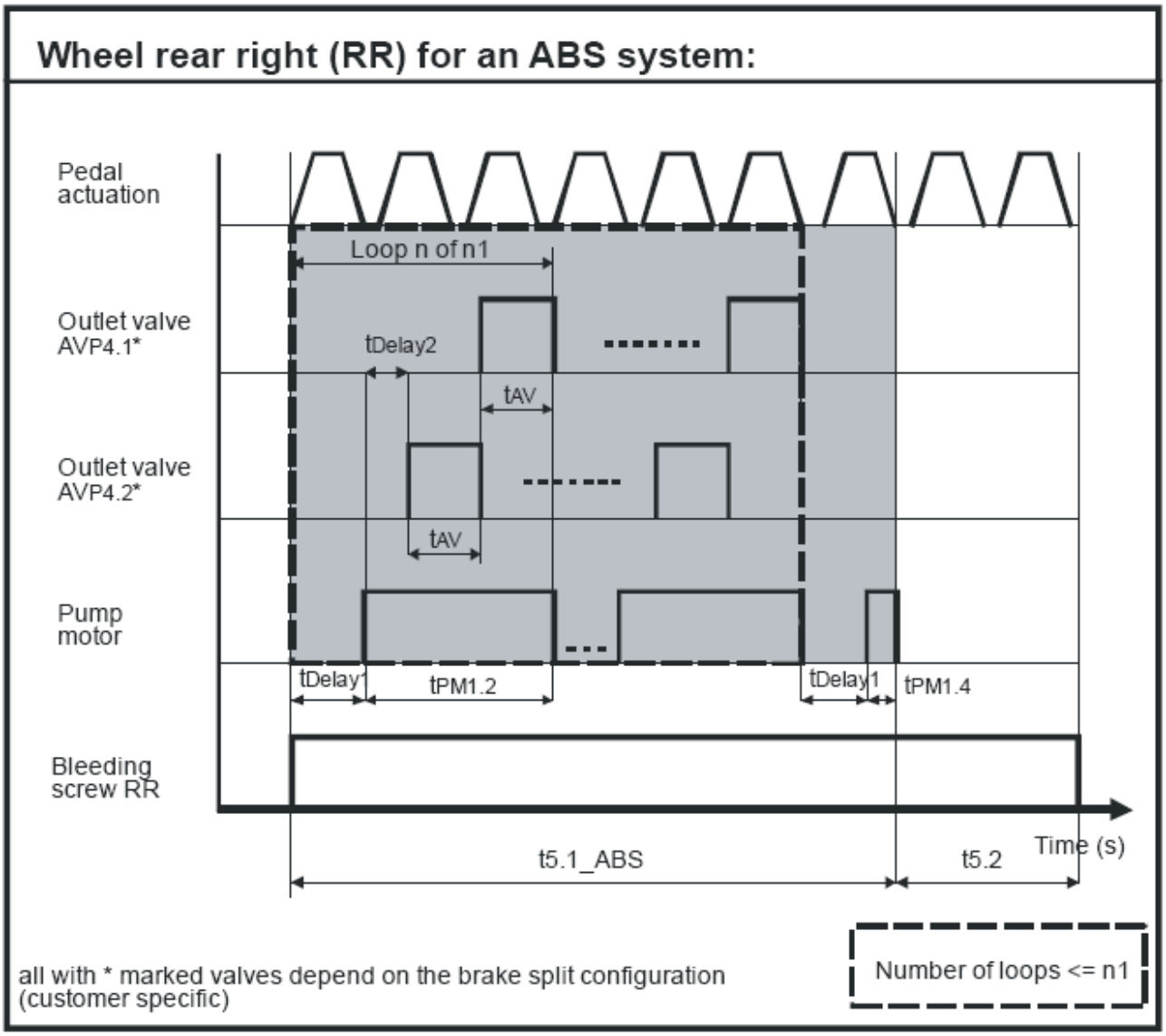

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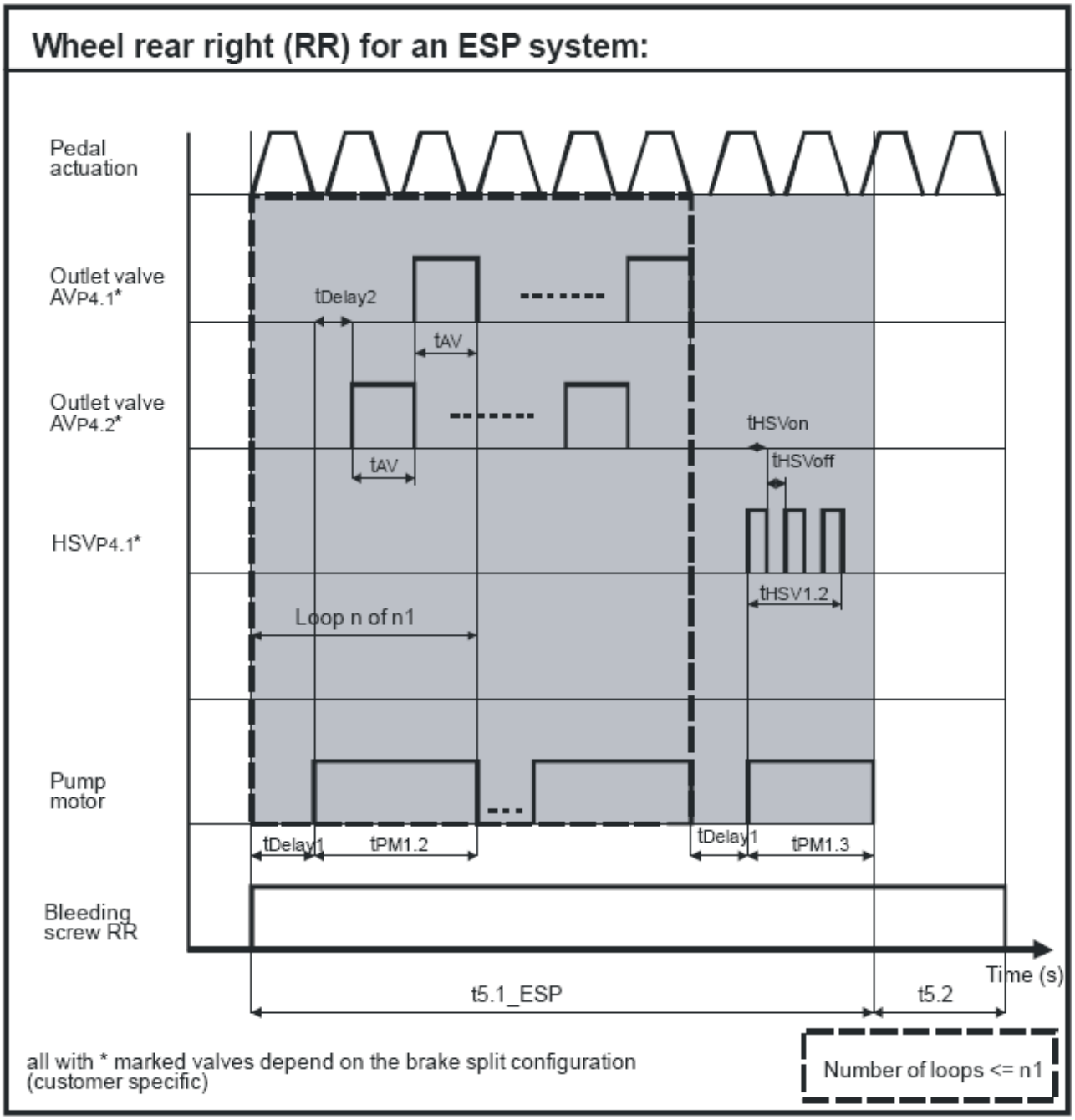
Diagram for wheel RR for a ESP system: ESP系统后右轮排气示意图：



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Diagnostic parameter: 诊断参数：

Parameter:	Physical values:
Valve activation time t_{AV1} 阀动作时间t_{AV1}	y s
Valve activation time $t_{HSV1.2}$ 阀动作时间$t_{HSV1.2}$	y s
Valve activation time t_{HSVon} 阀动作时间t_{HSVon}	y s
Valve activation time t_{HSVoff} 阀动作时间t_{HSVoff}	y s
Waiting time t_{Delay1} 等待时间t_{Delay1}	y s
Waiting time t_{Delay2} 等待时间t_{Delay2}	y s
Pump motor activation time $t_{PM1.2}$ 泵马达动作时间$t_{PM1.2}$	y s
Pump motor activation time $t_{PM1.3}$	y s
Pump motor activation time $t_{PM1.4}$	y s
Pump motor activation time $t_{PM1.5}$	y s
Number of loops n_1 循环次数n_1	y
Max. Phase duration $t_{5.1_ABS}$ 最长阶段持续时间$t_{4.1_ABS}$	y s
Max. Phase duration $t_{5.1_ESP}$ 最长阶段持续时间$t_{4.1_ESP}$	y s
Waiting time $t_{5.2}$ 等待时间$t_{4.2}$	x s

Notes: 注意：

The with x marked time value depends on the size of the base brake system and pressure p0. This time has to be determined by trails executed at the customer plant. The responsibility for these trials has the customer. All with y marked **time values** are to be inferred from the chapter **Diagnostic data (timing values)**.
数值为x的时间值取决于基础制动系统的尺寸和压力值p0。该时间必须在客户工厂通过试验获得，这些试验应由客户自行完成。数值为y的时间值可由“诊断数据（时间值）”相关章节得出。

4.11 Diagnosis step 11: Evaluation of the status of the valve activation 诊断步骤 11：阀动作状态评估

Details can be taken from the chapter 4.5. 具体请参考4.5小节。

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4.12 Diagnosis step 12: Evaluation of the fault code memory 诊断步骤 12：故障码内存评估

Evaluation of the fault code memory 故障码内存评估	
Diagnostic unit 诊断单元	ECU 电子控制单元
Evaluation of all relevant fault code entries after the process is finished. 在排气结束之后对所有相关故障码记录进行评估	Output of all stored fault codes 输出所有存储的故障码
More information about the chapter “Evaluation of the fault code memory” is to be inferred from the customer specific documentations. 关于“故障码内存评估”更多信息请参考客户规范文件。	

Notes: 注意：

To verify if all valves and the pump motor were correct activated in the diagnostic mode, the fault code memory has to be evaluated. Only relevant and actual faults have to be considered. The ABS/ESP ECU gives out the complete contents of the entire fault code memory. These data consists of the following information:

- Diagnostic trouble code (DTC)
- Status of DTC (actual / historic)

The diagnostic tester has to verify the complete outputted contents of the fault code memory. In case of relevant diagnostic trouble codes are stored, they have to be converted into a failure text and a fault protocol has to print out. The status information of a stored DTC shows if it is actual or historic.

为了确认是否所有阀和泵马达都在诊断模式里正确启动，必须评估故障码内存。只有与排气相关且真实的故障码才需要进行评估。ABS/ESP电子控制单元输出整个故障码内存的完整内容，包括的数据信息如下：

- 诊断故障码（DTC）
- DTC状态（真实/历史）

诊断设备必须对故障码内存输出的全部内容进行确认。如果存储了相关诊断故障码，这些故障码必须转换为故障文本并打印错误协议。存储的DTC状态信息显示该DTC为真实或历史DTC。

Only following faults are relevant for a correct executed repair bleeding process all other stored faults (e.g. missing calibration, missing variant code, etc.) can be ignored:
只有下列故障是与人工排气过程正确运行相关的，其他存储的故障（如缺乏校正、缺少变量编码等）都可以忽略。

Description of the diagnostic trouble code: 诊断故障码描述：	**Will be detected by evaluation of: 将在下列评估中发现：**
Outlet valves 出液阀	Status of valve activation 阀动作状态
Inlet valves 进液阀	
High pressure switching valves (only ESP systems) 高压开关阀（仅限ESP系统）	

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Power supply for the valves 阀电源	
Power supply for the pump motor 泵马达电源	Fault code memory 故障码内存
Pump motor 泵马达	

None relevant fault code stored ⇒ Continue
Relevant fault code stored ⇒ Abort, print out fault message
相关故障码未存储 ⇒ 继续
相关故障码存储 ⇒ 退出，打印错误信息

4.13 Write process bytes (optional) 写入过程字节（可选）

Write process bytes 写入过程字节	
Diagnostic unit 诊断单元	ECU 电子控制单元
Write process bytes 写入过程字节	Writes the process results into the EEPROM of the ABS/ESP system 将过程结果写入ABS/ESP系统的EEPROM
More information to the chapter “Write process bytes” is to be inferred from the customer specific documentations. 关于“写入过程字节”更多信息请参考客户规范文件。	

Notes: 注意：

All with **tbd** (to be defined) marked values are customer specific. With this optional function it is possible to write additional information about the repair bleed process into the EEPROM of the ABS/ESP system. The ABS/ESP ECU is delivered with predefined value (customer specific). This value is written into the ABS/ESP EEPROM by the Bosch plant. In case of none successful communication with the ABS/ESP ECU at the customer assembly plant, the delivery value is still stored. These conditions must be considered as a not successful respectively faulty repair bleed process. The offered EEPROM location to store this production information amounts up to two data bytes (customer specific). The definition and to be written process data is in responsibility of the customer. To do the right evaluation all defined values and data must be hand out to the corresponding suppliers (e.g. end of line equipment).

数值为**tbd**的所有值为客户指定。该可选功能允许在ABS/ESP系统的EEPROM中写入人工排气过程附加信息。ABS/ESP ECU在发送至客户时已由博世工厂在ABS/ESP EEPROM中预先写入了一个预定义值（由客户指定）。如果ABS/ESP ECU在客户装配工厂中通讯不成功，则该预定义值仍将存储在EEPROM中。这类情况必须被视为人工排气过程不成功或发生错误。EEPROM中有两个字节的地址位用于存储该附加信息（由客户指定）。客户负责过程字节的定义和写入。为了保证后续过程中能对过程字节进行正确评估，所有定义的数据都必须发送给相关联的设备供应商（如下线检测设备）。

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Example of writing two process bytes: 两个过程字节写入范例：

Repair bleed equipment writes the process bytes with detailed information: 人工排气设备写入过程字节（带详细信息）			
Fault description of the process byte 1: 过程字节1的错误描述	Fault description of the process byte 2: 过程字节2的错误描述	PB 1 data to be written (hex): 过程字节1	PB 2 data to be written (hex): 过程字节2
Fault during activation of the valves for the wheel FL 前左轮阀动作错误 Entire repair bleed process i.O. etc. 整个人工排气过程 i.O. etc.	Repair bleed facility 1 排气设备1 Repair bleed facility 2 排气设备2 etc.	tbd tbd etc.	tbd tbd etc.
Repair bleed equipment writes the process bytes without detailed information: 人工排气设备写入过程字节（无详细信息）			
Fault description of the process byte 1: 过程字节1的错误描述	Fault description of the process byte 2: 过程字节2的错误描述	PB 1 data to be written (hex): 过程字节1	PB 2 data to be written (hex): 过程字节2
Process ok 过程合格 Process nok 过程不合格	Default respectively de livery value 默认或预定义值	tbd tbd	tbd tbd

Write process bytes ok ⇒ **Continue**
Write process bytes nok. ⇒ **Print out fault message, abort**
写入过程字节合格 ⇒ 继续
写入过程字节不合格 ⇒ 打印错误信息，退出

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4.14 Diagnosis step 14: Clear fault code memory (optional) 诊断步骤 14：清除故障码（可选）

Clear fault code memory 清除故障码	
Diagnostic unit 诊断单元	ECU 电子控制单元
Clearing of the fault code memory 清除故障码内存	Fault code memory is erased. 故障码内存被清除
More information to the chapter “Clear fault code memory (optional)” is to be inferred from the customer specific documentations. 关于“清除故障码（可选）”更多信息请参考客户规范文件。	

Notes: 注意：
With this step the complete contents of the fault code memory is erased. This step is optional and can also be executed at a later process step.
该步骤将清除故障码内存中全部内容。该步骤作为可选步骤，也可以在后续的步骤中完成。

Clearing of the fault code memory successful ⇒ Continue
Clearing of the fault code memory not successful ⇒ Print out fault message, continue
清除故障码内存成功 ⇒ 继续
清除故障码内存不成功 ⇒ 打印错误信息，退出

4.15 Diagnosis step 15: End of diagnostics 诊断步骤 15：诊断结束

End of diagnostics 诊断结束	
Diagnostic unit 诊断单元	ECU 电子控制单元
Leaving of the customer specific diagnostic mode. 退出客户指定诊断模式	Leaves the customer specific diagnostic mode. 退出客户指定诊断模式
More information to the chapter “End of diagnostics” is to be inferred from the customer specific documentations. 关于“诊断结束”更多信息请参考客户规范文件。	

Notes: 注意：
With this step the customer specific diagnostic mode in finished respectively left. This step is required to change back to regular (normal) ABS/ESP control mode. In case of the valve relay is disabled caused by a detected fault a changing of the diagnostic mode is might not be possible (behaviour is customer specific).
该步骤将结束并退出客户指定诊断模式。通过该步骤，系统可以切换回常规ABS/ESP控制模式。如果检测到有故障导致阀继电器被禁用，切换诊断模式可能无法实现（由客户指定）。

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5 Failure handling 故障解决

Failure handling 故障解决	
Root cause of failure: 故障根本原因：	Failure message, failure handling: 故障信息，故障解决：
Bleeding device 排气设备	Failure message, mark vehicle with failure label → Repeat the complete repair bleeding at all four wheels 故障信息，在车辆上做故障标注 → 对所有车轮重复整个排气过程
Diagnostic unit 诊断单元	Failure message, mark vehicle with failure label → Repeat the complete repair bleeding at all four wheels 故障信息，在车辆上做故障标注 → 对所有车轮重复整个排气过程

Notes: 注意：

In case of any occurred failure the entire repair bleeding process has to be repeated at all four wheels.
如果发生任何故障，必须在所有四个车轮上重新进行整个人工排气过程。

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6 Data 数据

6.1 Process data (pressure and time values) 过程数据（压力和时间值）

Process step: 步骤	Parameter: 参数	Condi-tions: 状况	Physical value: 物理数值	Unit: 单位
1 to 8	Absolute pressure p0 绝对压力值p0	≥	3	Bar
		≤	tbd*	
	System pressure p1 系统压力值p1	≥	15	Bar
	Time t_{BP1} 时间t_{BP1}	~	3	Seconds
	Time t_{BP2} 时间t_{BP2}	~	0,5	

Notes: 注意：

All with **tbd** (to be defined) marked values are customer specific. The with x marked maximal allowed absolute pressure p0 depends on the maximal allowed pressure for the brake fluid reservoir. This value is customer specific.
数值为**tbd**（待定）的所有参数为客户指定。标有“*****”的最大允许绝对压力值p0取决于制动液储液壶最大允许压力。该值为客户指定。

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6.2 Diagnostic data (timing values) 诊断数据（时间值）

Process step: 步骤	Parameter: 参数	Condi-tions: 状况	Physical value: 物理数值	Unit: 单位
4	Phase duration $t_{2.1}$ 阶段持续时间$t_{2.1}$	=	25	Seconds
	Waiting period $t_{2.2}$ 等待时间$t_{2.2}$	=	tbd	
6	Phase duration $t_{3.1}$ 阶段持续时间$t_{3.1}$	=	25	
	Phase duration $t_{3.1_3K_ABS}$ 阶段持续时间$t_{3.1_3K_ABS}$	=	35	
	Waiting period $t_{3.2}$ 等待时间$t_{3.2}$	=	tbd	
8	Max. Phase duration $t_{4.1_ABS}$ 最长阶段持续时间$t_{4.1_ABS}$	≤	99*	
	Max. Phase duration $t_{4.1_ESP}$ 最长阶段持续时间$t_{4.1_ESP}$	≤	106*	
	Waiting period $t_{4.2}$ 等待时间$t_{4.2}$	=	tbd	
10	Max. Phase duration $t_{5.1_ABS}$ 最长阶段持续时间$t_{5.1_ABS}$	≤	79*	
	Max. Phase duration $t_{5.1_ESP}$ 最长阶段持续时间$t_{5.1_ESP}$	≤	86*	
	Waiting period $t_{5.2}$ 等待时间$t_{5.2}$	=	tbd	

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General valid data for following diagnosis steps: 诊断步骤对应的常规有效数据：					
4, 6, 8 and 10	Number of loops n_1 循环次数n_1	*min.*	≥	3	---
		recommended	=	4	
		max.	≤	5	
	Valve activation time t_{AVon}		=	1	Seconds
	Valve activation time t_{AVoff}		=	1	
	Valve activation time t_{AV0}		=	20	
	Valve activation time t_{AV1}		=	5	
	Valve activation time t_{AV2}		=	15	
	Valve activation time t_{AV3}		=	10	
	Valve activation time $t_{HSV1.1}$		=	10	
	Valve activation time $t_{HSV1.2}$		=	7	
	Valve activation time t_{HSVon}		=	1	
	Valve activation time t_{HSVoff}		=	0,5	
	Waiting time t_{Delay1}		=	5	
	Waiting time t_{Delay2}		=	3	
	Pump motor activation time $t_{PM1.1}$		=	10	
	Pump motor activation time $t_{PM1.2}$		=	13	
	Pump motor activation time $t_{PM1.3}$		=	9	
	Pump motor activation time $t_{PM1.4}$		=	2	
	Pump motor activation time $t_{PM1.5}$		=	8	

Notes: 注意：

All with **tbd** (to be defined) marked values are customer specific.
The with x marked maximal phase durations **t4.1_ABS, t4.1_ESP, t5.1_ABS** and **t5.1_ESP** are the values for the recommended numbers of loops.
数值为**tbd**（待定）的所有参数为客户指定。
标有 **"*"**的最长阶段持续时间**t4.1_ABS, t4.1_ESP, t5.1_ABS**和**t5.1_ESP**适用于推荐的循环次数。

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6.3 Hydraulic data 液压数据

6.3.1 Table of valve assignment depending on the brake circuit configuration 不同制动回路分布类型下的阀分配表

Diagnosis step: 诊断步骤	Valve activation parame-ter: 阀动作参数	Diagonal brake split configuration (X): X型制动回路分布		Front/rear brake split configuration (II) II型制动回路分布		
		4- channel				3- channel
		ABS	ESP	ABS	ESP	ABS
4	$AV_{P1.1}$	FR		RR		---
6	$AV_{P2.1}$	RR		FR		RA
	$AV_{P2.2}$	---		---		FR
	$EV_{P2.1}$	---		---		FL
	$EV_{P2.2}$	---		---		FR
8	$AV_{P3.1}$	RL		FL		FL
	$AV_{P3.2}$	FL		RL		FR
	$AV_{P3.3}$	FR		FR		---
	$EV_{P3.1}$	FR		FR		RA
	$EV_{P3.2}$	RR		RR		FR
	$HSV_{P3.1}$	---	FL/RR	---	RL/RR	---
	$HSV_{P3.2}$	---	FR/RL	---	FL/FR	---
10	$AV_{P4.1}$	FL		RL		RA
	$AV_{P4.2}$	RR		RR		---
	$HSV_{P4.1}$	---	FL/RR	---	RL/RR	---

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6.3.2 Volume and valve orifice data respectively pump element dry run time 容积、阀截流口孔径以及泵单元干转时间

Volume and valve orifice data respectively pump element dry run time: 容积、阀截流口孔径以及泵单元干转时间：	ABS 9	ESP 9
Volume of the primary circuit per circuit (cm3): 第一回路每条油路容积（cm3）	tbd	tbd
Volume of the secondary circuit per circuit (cm3): 第二回路每条油路容积（cm3）	tbd	tbd
Volume of the priming circuit per circuit (cm3): 第一回路每条油路容积（cm3）		tbd
Dead space of the pump elements per circuit (cm3): 泵单元每条油路盲区（cm3）	tbd	tbd
Inlet valve orifice diameter RA (mm): 后轴进液阀截流口孔径（mm）	customer specific 客户指定	
Inlet valve orifice diameter FA (mm): 前轴进液阀截流口孔径（mm）	customer specific 客户指定	
Outlet valve orifice diameter RA (mm): 后轴出液阀截流口孔径（mm）	customer specific 客户指定	
Outlet valve orifice diameter FA (mm): 前轴出液阀截流口孔径（mm）	customer specific 客户指定	
Maximal Pump element dry run time t_{PEdry} (s): 泵单元最长干转时间t_{PEdry}（s）	≤ 60	

Notes: 注意：

All mentioned volume data are rounded values. The used orifices are project and customer specific.
所有上述容积数据都是圆整值。采用的孔径大小为项目/客户指定。

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6.4 Electrical data 电气数据

6.4.1 Information to the calculated current consumption of an ABS/ESP system ABS/ESP 系统的电流消耗计算值

Calculated current consumption at a 12V voltage supply 12V电压电源下的电流消耗值				
Component:		**Component:**	**Compo-nent:**	**Component:**
ECU:		**≤ 0,5**	**A**	
AV coil (7,3 Ω): 出液阀线圈		**1,64**	**A**	
EV coil (7,3 Ω): 进液阀线圈		**1,23**	**A**	**Valve protection measures active! For more details please refer the chapter Valve protection measures at generation 9.** 阀保护措施启动！具体请参考**9代系统阀保护措施**。
HSV coil (3,6 Ω): 高压开关阀线圈		**3,3/0,97**	**A**	
Max. total valve current consumption: 阀组电流消耗最大值	**ABS:**	**6,02**	**A**	**Only valid if the defined valve activation ratio and profile is kept (worst case at an ABS: 2xAV+2xEV, at an ESP: 2xEV+2xHSV to the same time) at each wheel.** 仅在按照规定的阀动作占空比和曲线图运行时有效（**ABS**最坏情况：同一时间**2xAV+2xEV**；**ESP**最坏情况：同一时间**2xEV+2xHSV**）。
	6,40			
PM start up current I_{on} 泵马达启动电流I_{on}		**≤ 145**	**A**	**Depending on the used pump motor.** 由所用泵马达型号决定。
PM start up current time t_{on} 泵马达启动电流持续时间		**≤ 20**	**ms**	
PM constant current I_{on_const} 泵马达稳定电流		**≤ 15**	**A**	

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6.4.2 Specification of the voltage supply (optional) 电压供应规格（可选）

Specification voltage supply 电压供应规格		
Parameter:	**Physical value:**	**Remarks:**
Range of voltage [V]: 电压范围	11,5 ≤ power supply unit ≤ 14	Allowed voltage range measured at the ECU. It must be guaranteed that the voltage never drops below 10V (e.g. buffering of the voltage with a condenser required)! 电压允许范围在ECU处测得。电压必须保证不低于10V（例如使用电容器作为电压缓冲器）
Current [A]: 电流	≥ 20	To supply the vehicle battery. 供应汽车蓄电池
	≥ 145	To supply the complete vehicle (no battery installed). 供应整车（蓄电池未安装）

Notes: 注意：

The repair bleed equipment respectively its power supply hast to meet the above described requirements. Independent of the length of the wires between the power supply and the ABS/ESP ECU and the possible caused voltage drop the specified electrical values must be kept.
人工排气设备及其电源供应必须满足上述要求。电源与 ABS/ESP ECU 之间线束长度可能引起一定的电压降，但对以上规定的电气数据不构成影响。

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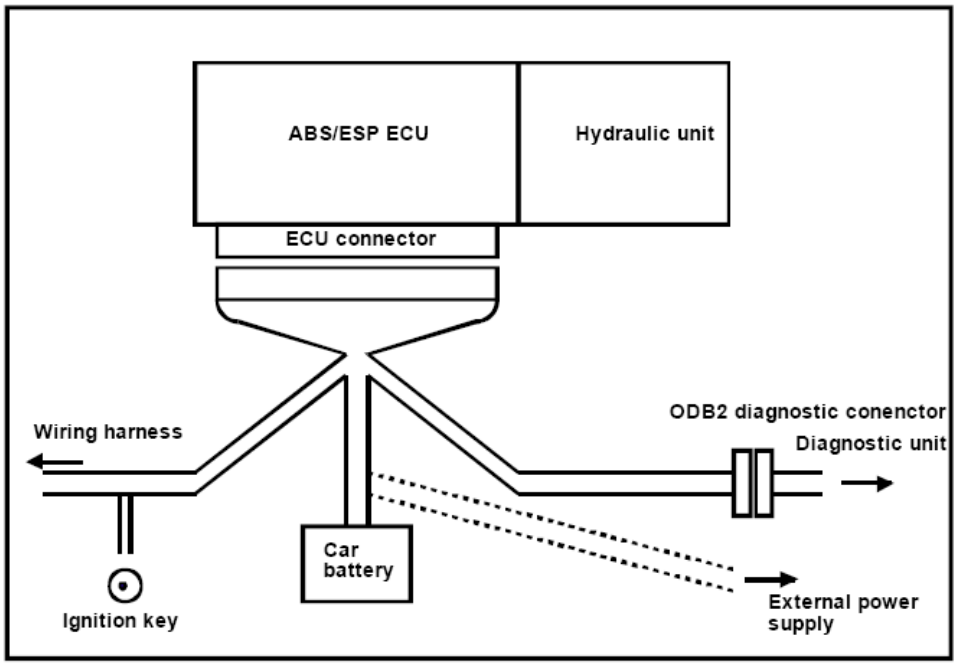
6.4.3 Connection at the diagnostic connector (OBD2) within the vehicle 连接车内诊断（OBD2）接口

Handling 操作	
Operator 操作者	Remarks: 备注
☞ Adapt diagnostic connector 连接诊断接头 ☞ Adapt power supply 连接电源 ☞ Switch ignition key to position on 打开点火开关	Either in the passenger or engine compartment. External power supply is required either to supply the car battery or the ABS/ESP ECU itself (e.g. car battery is not installed prior this process step). 在驾驶室或发动机舱内。需要外接电源为汽车蓄电池或ABS/ESP ECU 供应电能（例如汽车蓄电池此时尚未安装）。 To switch on and initialize the ABS/ESP system. 点火并初始化ABS/ESP系统。

Notes: 注意：
The length of the diagnostic connection wires should be covered and should not exceed 10 meters.
诊断连接线外部应当裹好，线缆长度不能超过10米。



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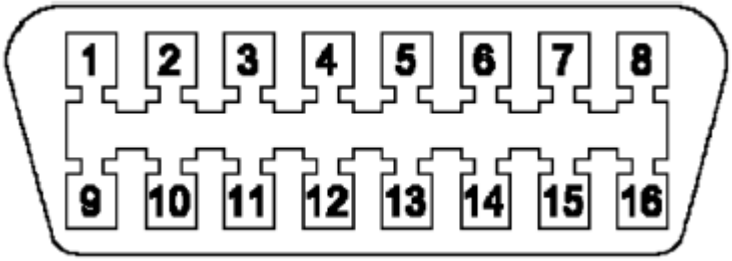
6.4.4 Specification of the diagnostic OBD2 connector within the passenger cell 驾驶舱内诊断（OBD2）接口规范

Specification of the OBD2 diagnostic connector (female): 驾驶舱内诊断（OBD2）接口规范	
	
Description of the Pin: 针脚描述	**Pin- No.: 针脚编号**
Customer specific 客户指定	**1-3**
Vehicle ground 阀地线	**4**
Signal ground 信号地线	**5**
CAN-High	**6**
ISO 9141 K-Line	**7**
Customer specific 客户指定	**8-13**
CAN-Low	**14**
Customer specific 客户指定	**15**
+12V power supply +12V电源	**16**

Notes: 注意：

The used connector has to be an industrial one. It is not recommended to use a simple standard connector, because with a standard connector only limited safe connections are possible. To ensure a safe connection a lot of times at a serial production an industrial connector with golden terminals should is required.
All customer specific information and to be used diagnostic interface have to be taken from the project specific documents.
连接器必须是工业型连接器。我们不推荐使用简单的标准型连接器，因为标准型连接器无法完全保证安全连接。为了确保批量生产中能进行多次安全连接，必须采用带镀金导电触片的工业型连接器。
所有客户规范信息以及采用的诊断接口都必须参考项目规范文件。

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7 Valve protection measures at generation 9　9 代系统的阀保护措施

	ABS	ESP
Valve:	**Diagnostic mode 诊断模式**	
EV **7,3 Ω**	**If UVR > 9V, than lowering of the duty cycle to Ueff = 9V; If valve activation value > 0%; than start the up/down counter and count with a ratio up/down = 2/1; limited to 120s; for each EV valve a separate counter is used** **如果UVR > 9V，则降低占空比至Ueff = 9V；如果阀启动值 > 0%，则上/下计数器以2/1的上/下比开始计数，限时120s；每个进液阀都有单独的计数器。**	
AV **7,3 Ω**	**If valve activation value > 0% (via customer diagnosis only full time =100% activation possible); than start the up/down counter and count with a ratio up/down = 2/1; limited to 30s; for each AV valve a separate counter is used** **如果阀启动值 > 0%（在客户诊断中只允许100%全时启动），则上/下计数器以2/1的上/下比开始计数，限时30s；每个出液阀都有单独的计数器。**	
HSV **3,3 Ω**	**---**	**20ms with 100% duty cycle, than lowering of the duty cycle to $U_{effective}$ = 3,5V and cyclic all 200ms for 20ms with 100%** **以100%的占空比启动20ms，然后降低占空比至$U_{effective}$ = 3,5V；每200ms中20ms为100%占空比，循环进行。**
USV **7,3 Ω**	**---**	**Not activated 未启动**
USV **4,3 Ω**		

Notes: 注意：

All valve protection measurements are active in the diagnostic mode. These measurements can not be switched off.
在诊断模式下，所有阀保护措施都被激活，无法关闭。

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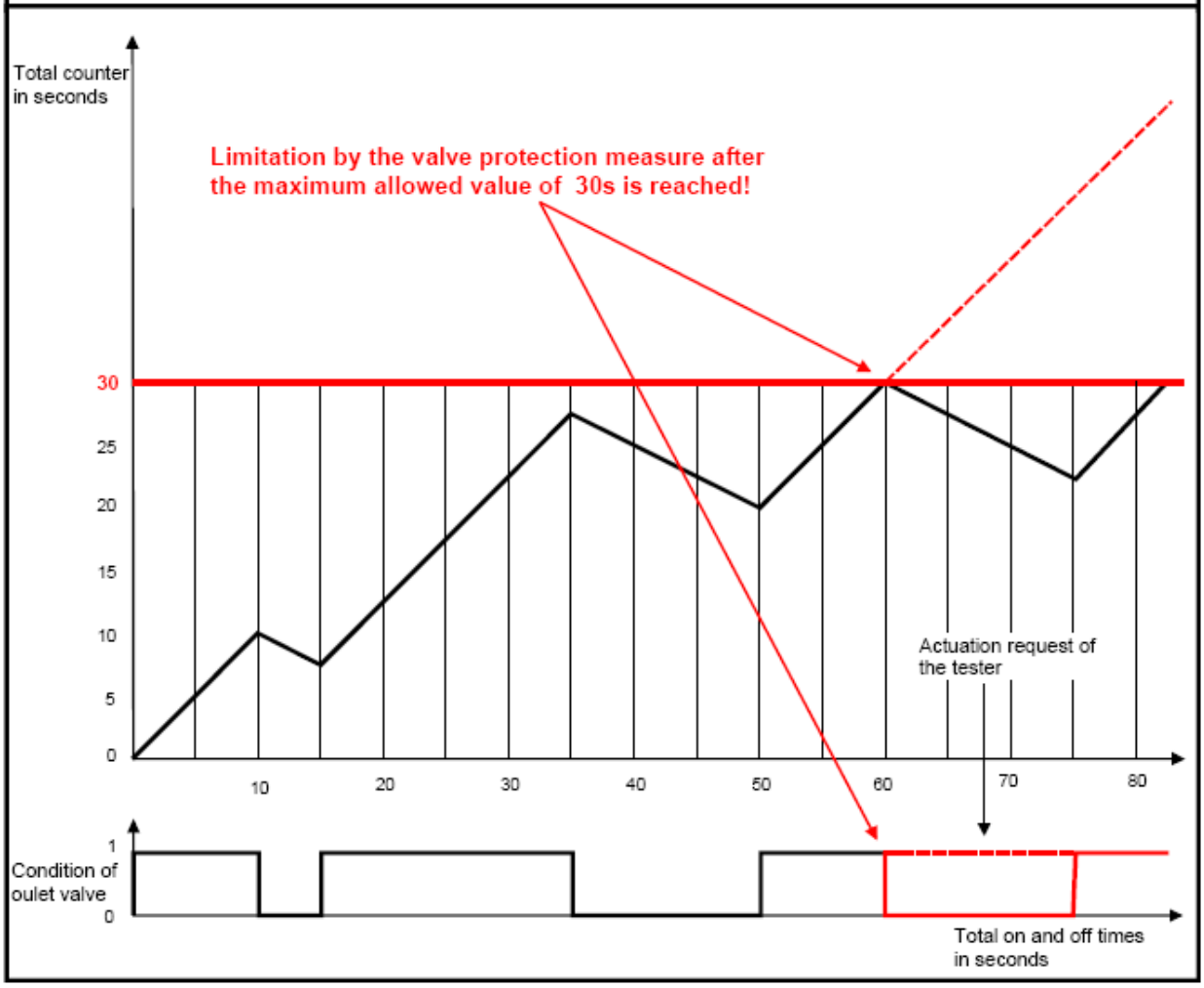
Gen 9: Valve protection measure: Up/down counter with a ratio of = 2/1

Example: Outlet valve with a maximum limitation to 30 seconds



Notes:注意：

A forced limitation by the valve protection measurement is active as long as the valve coils and the ECU are protected against overheating. First after a corresponding cooling time the valves can be switched on again. To ensure the important coil respectively ECU cooling time after a executed valve actuation via diagnosis in a previous ignition cycle (after a hardware or software reset), the actual coil resistance values are measured and used after each power up to initialise the start up value of the up/down counters. Caused by different reasons the described valve protection measurements cannot protect the valve coils and the ECU with 100% safeness. That's the reason why the customer is responsible to ensure a correct valve activation sequence!

阀保护措施强制保护阀电磁线圈与 ECU 免于过热。当冷却时间足够后阀组方能再度启动。

有时前一个点火循环中利用诊断功能启动了阀，而再次点火后硬件和软件都重启了。为了保证阀电磁线圈与 ECU 有足够冷却时间，可以通过测量阀线圈的电阻值来判断上/下计数器的合理初始值。出于多种原因，即便是阀保护措施也无法保证阀电磁线圈和 ECU100%安全。因此，客户有责任确保阀启动序列正确！

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8 Equipment information 设备信息

Diagnostic information 诊断信息

A diagnostic unit is required to activate the valves and the pump motor. This is required to evacuate and fill the secondary circuits of the ABS/ESP system. To avoid problems with the communication (especially in case of k-line diagnosis) between the diagnostic unit and ABS/ESP ECU, the diagnostic unit should be located as close as possible at the vehicle.

为了对第二回路进行抽真空加注，诊断单元应启动ABS/ESP系统的阀和泵马达。为了避免诊断单元和ABS/ESP ECU之间的通讯问题（特别是使用k线诊断时），诊断仪应当尽量靠近车辆。**Process bytes 过程字节**

The final definition of the process byte data which should be written at the repair bleed process must be defined by the customer.

过程字节必须由客户定义并在人工排气结束后写入。

Editor menu 编辑菜单

The ECU identification data might be changed by software update during a running serial production. To have the possibility to add these new ECU data at the repair bleed equipment, the editor menu must be designed in a way that a complement of these missing new ECU identification data is easily possible for the operator.

在量产过程中，ECU识别数据可能随软件升级而发生变更。为了方便操作员将这些新数据添加进人工排气设备，编辑菜单应当设计得简单易用。

Test protocol 测试协议

In case of an ok test result, a print out (”test passed”) is made.
In case of a nok test result, a failure protocol must be printed out.
This protocol must include all technical information about the failed step in order to allow a better failure analysis.
An entire test protocol of all process step results must be available on request.
如果测试合格，打印“测试通过”。
如果测试不合格，打印错误协议。
为了更好的分析问题，该协议必须包括错误步骤在内的所有技术信息。
测试协议应当包括所有步骤的结果。

Vehicle conditions 车辆条件

The vehicle battery must be mounted and properly charged in order to allow a sufficient power supply to the ABS/ESP system. An external power supply should only be used to charge the battery.

蓄电池应妥善安装并充电以保证ABS/ESP系统能得到充足的电力供应；同时外接电源的作用必须仅限于对蓄电池充电。

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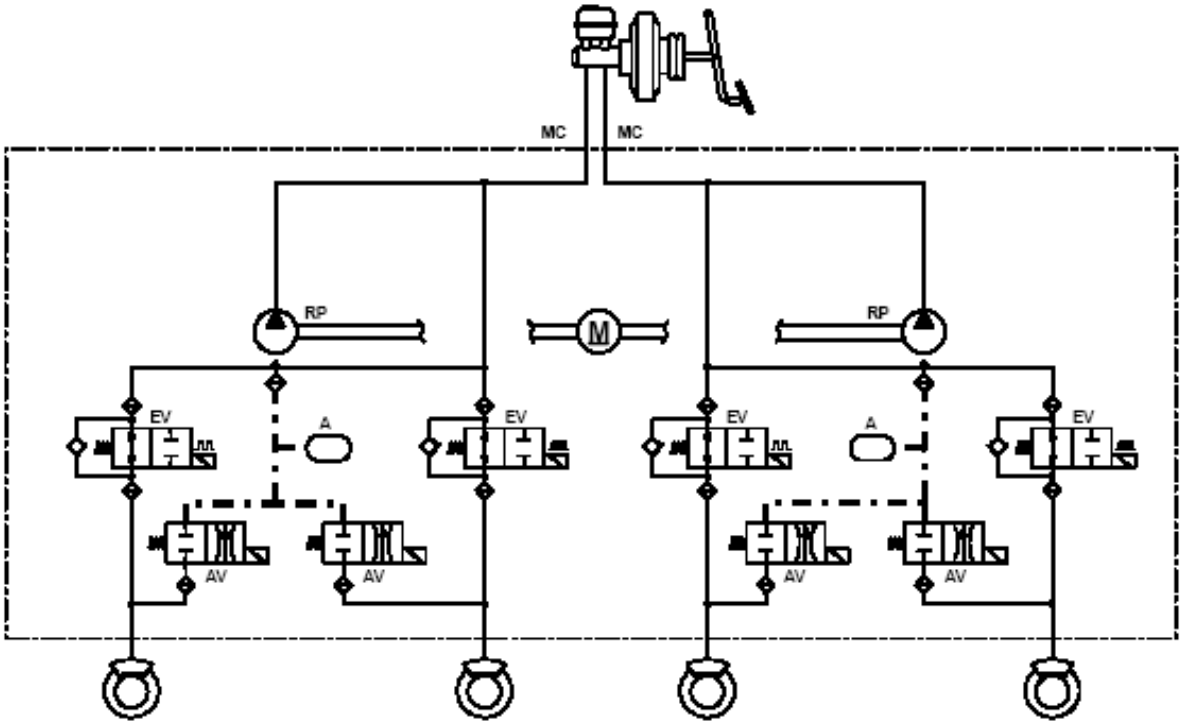
9 Attachments 附件

9.1 Hydraulic Diagrams 液压回路图

9.1.1 ABS 9 (4-channel) – independent of the brake split configuration

ABS 9（4通道）– 适用于所有回路布置方案



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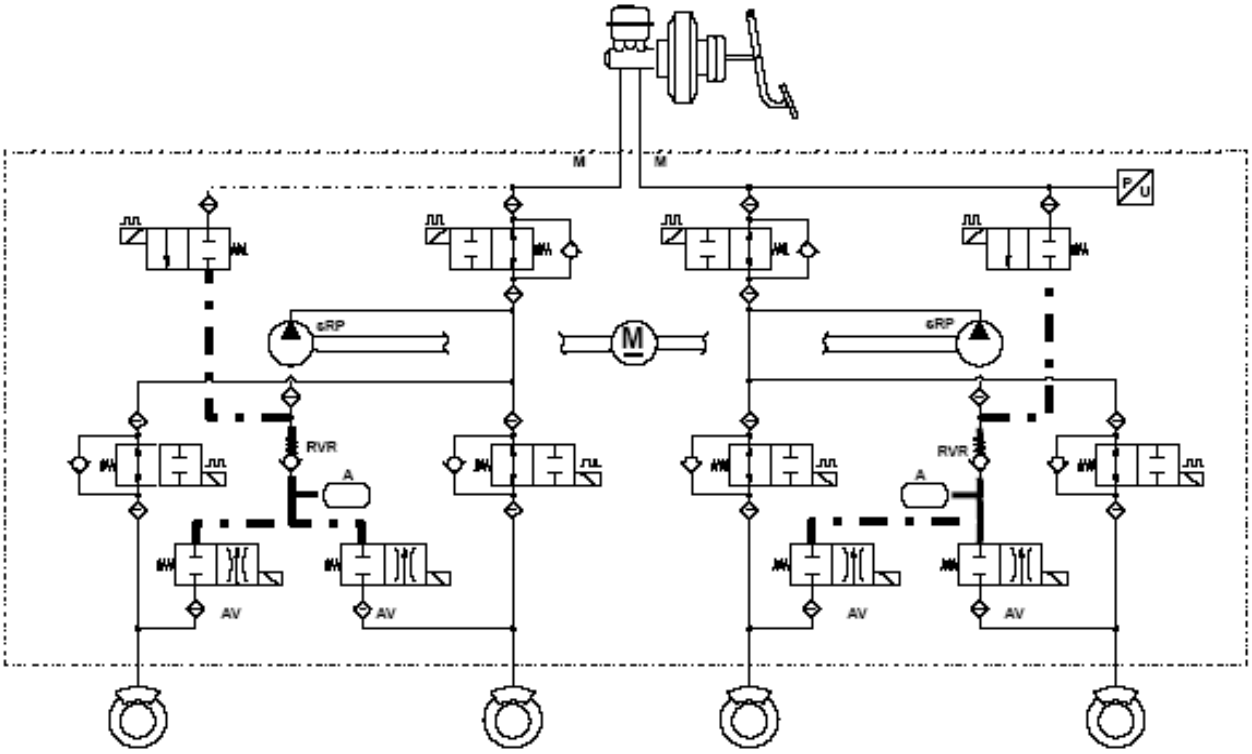
9.1.2 ESP 9 - independent of the brake split configuration

ESP 9（4通道）– 适用于所有回路布置方案



The fat marked dotted lines within the ABS respectively ESP hydraulic unit shows the area of the secondary and priming circuit.
ABS/ESP液压单元中加粗虚线部分表示第二回路和第一回路。

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